


7

STATISTICS AND PROBABILITY

# INVESTIGATING DATA

The Bureau of Meteorology collects weather data across Australia, including rainfall and temperatures. Polling organisations like Roy Morgan and Galaxy Research collect data on people's opinions on a wide range of issues. In NSW, the Bureau of Crime Statistics and Research (BOCSAR) collects data on crimes in different areas of the state. Health departments collect data on medical issues and health care. In all cases the data is analysed to evaluate decisions already made and to inform decision making in the future.



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## Chapter outline

| | Working mathematically | | | | |
|---|---|---|---|---|---|
| **7.01** Column, sector and divided bar graphs | U | F | | R | C |
| **7.02** Line graphs, pictograms and infographics | U | F | | R | C |
| **7.03** Types of data | U | F | | R | C |
| **7.04** Census vs sample | U | F | | R | C |
| **7.05** The mean and mode | U | F | | R | C |
| **7.06** The median and range | U | F | PS | R | C |
| **7.07** Analysing frequency tables | U | F | | | |
| **7.08** Dot plots and stem-and-leaf plots | U | F | PS | R | C |
| **7.09** Frequency histograms and polygons | U | F | PS | R | |
| **7.10** Choosing the best graph | U | F | | R | C |
| **7.11** The shape of data | U | F | | R | C |
| **7.12** Comparing data | U | F | PS | R | C |

## Wordbank

**bias** Something unwanted that causes a sample to not truly represent the population

**categorical data** Non-numerical data that can be classified into categories (for example, people's hair colour, or their favourite football team)

**frequency** How often a value appears in a data set

**frequency histogram** A special column graph that shows the frequencies of numerical data

**frequency polygon** A special line graph that shows the frequencies of numerical data

**infographic** A combination of graphs, images and words to describe information in a clear and interesting way

**measure of centre** An average, middle or typical value of a data set; a general name for the mean, median or mode

**numerical data** Data that can be measured and counted using values (for example people's height, or the number of devices they own)

**skewed** Twisted to one side, the shape of data that is not symmetrical

**Quiz**
Wordbank 7

**U** = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

## Videos (12):

**SkillCheck** Statistics

**7.03** Types of data

**7.06** The mean, mode, median and range
• The mode, median and mean

**7.07** Analysing frequency tables
• Statistics from a frequency table

**7.08** Analysing dot plots • Dot plots
• Stem-and-leaf plots 1
• Stem-and-leaf plots 2
• Back-to-back stem-and-leaf plots

**7.11** The shape of a frequency distribution

## *Twig* videos (8):

**7.01** Distorted graphs: Heat wave
• Nightingale's diagram

**7.04** Counting crowds • Can fish oil make you smarter? • Can you trust your IQ? • The wrong guy won

**7.05** Most popular pet

**7.06** Average Joe

## *PhET* interactive (1):

**7.06** Centre and variability

## Quizzes (5):

- Wordbank 7
- SkillCheck 7
- Mental skills 7
- Language of maths 7
- Test yourself 7

## Skillsheets (3):

**7.01** Displaying data

**7.05** Statistical measures

**7.06** Statistical measures
• Reading linear scales

## Worksheets (10):

**7.01** Every picture tells a story
• Frequency tables

**7.06** Mean, median, mode

**7.07** The mean and *fx* tables

**7.08** Stem-and-leaf plots • What data?

**7.09** Analysing data

**7.12** Analysing data • Comparing word lengths • Investigating young drivers

**Mind map** Investigating data

## Puzzles (8):

**7.01** Where all the cars are red

**7.03** Data match-up

**7.05** Looking for gold

**7.06** Ranges and averages • Data puzzles
• Mode, median and mean

**7.09** Statistical match-up • Highway elephants

**7.11** Statistical match-up

**Language of maths** Data crossword

## Technology (6):

**7.01** Creating a graph

**7.06** Students' marks • Finding the mean, median, mode and range
• Statistical measures calculator

**7.12** Population polygons
• Daily rainfall

## Presentation (1):

**7.08** Stem-and-leaf plots

## Spreadsheet (1):

**7.06** Statistical measures calculator

Nelson MindTap

To access resources above, visit
**cengage.com.au/nelsonmindtap**

## In this chapter you will:

- interpret and draw column graphs, sector graphs, divided bar graphs, line graphs, pictograms, infographics, dot plots and stem-and-leaf plots
- organise data into a frequency table
- identify categorical and numerical data, sort categorical data into ordinal and nominal data, and sort numerical data into discrete and continuous data
- identify ways of collecting data, including census, surveys, sampling and observation

- calculate and interpret the mean, median, mode and range of a set of data, including those represented on a graph or plot
- interpret and draw frequency histograms and polygons
- investigate the effect of individual data values, including outliers, on the mean and median
- identify the shape of a set of data

## SkillCheck ANSWERS ON P. 590





Quiz
SkillCheck 7



Video
Statistics

**1** A group of school students visiting Japan have the following ages.

14 13 12 15 13 14 12 14 15 14 14 13 13

Show this information on a dot plot.

**2** The masses (in kilograms) of some hospital patients are:

68 59 63 68 54 48 49 64 47 48 59 68 30

**a** Show this data on an ordered stem-and-leaf plot.

**b** What is the outlier?

**3** A group of people was surveyed about the number of phone calls they made on a Saturday night. The results are shown below.

6 4 0 3 1 2 2 3 5 4 3 3

4 1 0 6 6 3 4 3 1 5 4 3

**a** Draw a column graph for this data.

**b** How many people were surveyed?

**c** What was the most common number of phone calls?

**d** What was the least common number of phone calls?

**e** What was the highest number of phone calls?

**4** This back-to-back stem-and-leaf plot shows the points scored for and against the Koala Rugby team in each match, at one stage in the season.

| For | | Against |
|---|---|---|
| 4 | 0 | 6 |
| 8 4 | 1 | 4 6 6 |
| 2 2 0 0 | 2 | 2 9 9 |
| | 3 | 0 1 |
| 8 1 | 4 | 7 |
| 3 | 5 | |

**a** What is the highest number of points scored by the Koalas in one game?

**b** How many matches have the Koalas played so far?

**c** How many points have been scored against the Koalas?

# 7.01 Column, sector and divided bar graphs

**Skillsheet**
Displaying data

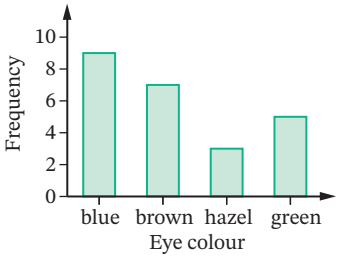
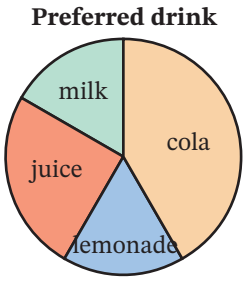
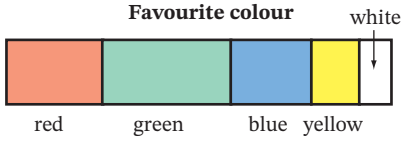
**Worksheet**
Every picture tells a story

**Puzzle**
Where all the cars are red

**Videos**
Distorted graphs: Heat wave

Nightingale's diagram

**Data** is a collection of facts or raw information. Data can be organised and displayed in a variety of ways so that we can **analyse** (study) it in more detail. This table shows 3 examples of statistical graphs.

| Graph | Description |
|---|---|
| Column graph (or bar chart)<br>  | • Mostly used for data that are in categories<br>• Uses columns to show the quantity or frequency of each category<br>• The horizontal axis (going across) shows the categories<br>• The vertical axis (going up) shows the frequency |
| Sector graph (or pie chart)<br>  | • Shows parts of a whole in circle form<br>• Uses sectors of a circle to show sizes of parts |
| Divided bar graph<br>  | • Shows parts of a whole in rectangular form<br>• Uses sections of a rectangle to show sizes of parts |

## Example 1

A sample of 80 people was surveyed on their favourite TV sport, and the results are shown.

| TV sport | Australian football | Cricket | Motor racing | Rugby league | Rugby union | Football |
|---|---|---|---|---|---|---|
| Frequency | 13 | 16 | 8 | 19 | 10 | 14 |

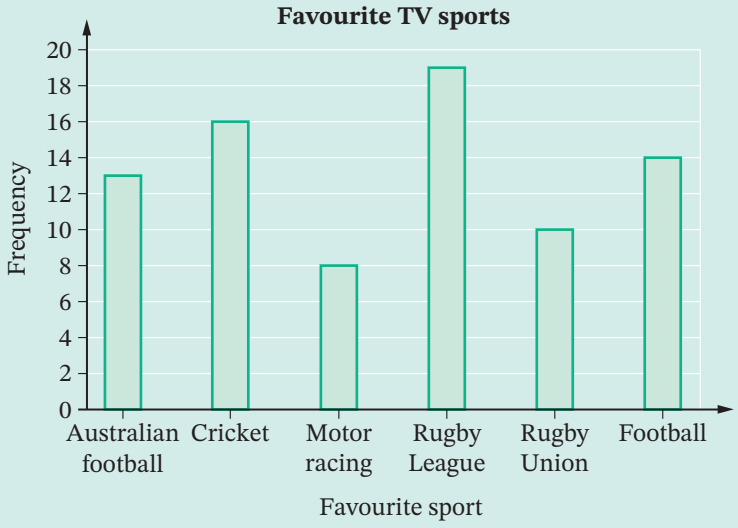
Display this data on a:

**a** column graph **b** sector graph **c** divided bar graph


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## Solution

**a** The highest frequency is 19 (for rugby league), so use a scale of 1 unit = 2 on the 'Frequency' axis. Label both axes, draw a column for each category, then label the graph.




**b** Draw a large circle, then calculate the angle sizes for each sector using fractions of 360° (because there are 360° in a revolution).

Total number of people in sample = 80.

Sector angles:

Australian football: $\frac{13}{80} \times 360° = 58.5°$

Cricket: $\frac{16}{80} \times 360° = 72°$

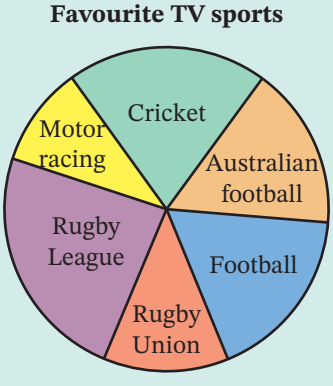
Motor racing: $\frac{8}{80} \times 360° = 36°$

Rugby League: $\frac{19}{80} \times 360° = 85.5°$

Rugby Union: $\frac{10}{80} \times 360° = 45°$

Football: $\frac{14}{80} \times 360° = 63°$

Construct the sector graph with the calculated angles, then label the sectors and the graph.



**c** Draw a rectangle whose length will represent 80, the total frequency (the whole), for example, 12 cm. Then calculate the length of each section using fractions of 12 cm.

Section sizes:

Australian football: $\frac{13}{80} \times 12\text{ cm} = 1.95\text{ cm}$

Cricket: $\frac{16}{80} \times 12\text{ cm} = 2.4\text{ cm}$

Motor racing: $\frac{8}{80} \times 12\text{ cm} = 1.2\text{ cm}$

Rugby League: $\frac{19}{80} \times 12\text{ cm} = 2.85\text{ cm}$

Rugby Union: $\frac{10}{80} \times 12\text{ cm} = 1.5\text{ cm}$

Football: $\frac{14}{80} \times 12\text{ cm} = 2.1\text{ cm}$

Construct the divided bar graph with the calculated section lengths, then label the sections and the graph.

**Favourite TV sports**




Worksheet
Frequency tables

## Frequency tables

A **frequency table** (or **frequency distribution table**) can be used to efficiently organise and count a large set of numerical data values. The **frequency** of a value is the number of times that value appears in the set of data.

### Example 2

20 people were surveyed about the number of cars in their household.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 1 | 2 | 0 | 3 | 2 | 1 | 1 | 4 |
| 1 | 1 | 1 | 2 | 2 | 0 | 3 | 2 | 1 | 2 |

Complete a frequency table for the data.

#### Solution

Draw a table with columns for Score, Tally and Frequency, as shown next page.
The lowest score is 0 and the highest is 4, so list the numbers 0 to 4 in the Score column.

For each value in the data set, draw a tally mark in the tally column. Tally means 'to count', and we are counting how often each value occurs.

The first value is a 2, so mark a '|' in the 'Tally' column beside 2. The second value is another 2, so mark another '|' there. Continue until you have counted all of the values.

| Score | Tally | Frequency |
|---|---|---|
| 0 | II | |
| 1 | 𝍸 II | 𝍸 means a tally of 5. |
| 2 | 𝍸 III | |
| 3 | II | |
| 4 | I | |
| Total | | |

For each value, count up the tally marks and write the answer in the 'Frequency' column. Then write the total in the final row and check that it is equal to the total number of values.

| Score | Tally | Frequency |
|---|---|---|
| 0 | II | 2 |
| 1 | IIII II | 7 |
| 2 | 𝍸 III | 8 |
| 3 | II | 2 |
| 4 | I | 1 |
| Total | | 20 |



## EXERCISE 7.01 ANSWERS ON P. 590

## Column, sector and divided bar graphs U F R C

**1** A class of 30 PE students was surveyed about the day of the week that each student trained for their chosen sport.

EXAMPLE 1

| Day | Monday | Tuesday | Wednesday | Thursday | Friday |
|---|---|---|---|---|---|
| Number of students | 3 | 13 | 4 | 7 | 3 |

Display this data in a sector graph. Label the graph. C

**2** The colours of the jelly beans in a jar were counted.

| Colour | Red | Black | Purple | Pink | Orange | White |
|---|---|---|---|---|---|---|
| Frequency | 11 | 5 | 2 | 4 | 6 | 8 |

Display this data as a column graph, using an appropriate scale. Label the graph and both axes. C



Shutterstock.com/Hayati Kayhan

**3** A sample of men were surveyed on the number of hours they slept the previous night.

| Number of hours | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|
| Frequency | 3 | 9 | 11 | 6 | 5 | 2 |

Display this data as a divided bar graph, 9 cm long. Label the graph. C

**4** This table shows the number of kilolitres of water used by a household in different areas over a year.

| Area | Garden | Shower | Toilet | Kitchen |
|---|---|---|---|---|
| Water usage (kL) | 90 | 520 | 140 | 50 |

If this information is shown in a divided bar graph of length 10 cm, which of the following is the length of the section representing 'Shower'? Select the correct answer **A**, **B**, **C** or **D**. R

**A** 1.5 cm **B** 1.9 cm **C** 5.2 cm **D** 6.5 cm

**5** The results of a spelling quiz (out of 10) for a class of students are as shown. C

EXAMPLE 2

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 7 | 7 | 8 | 9 | 7 | 5 | 4 | 8 | 9 |
| 9 | 8 | 10 | 8 | 7 | 5 | 5 | 4 | 1 |
| 8 | 8 | 9 | 7 | 7 | 4 | 8 | 3 | 1 |

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**a** Complete a frequency table for this data.

**b** What was the highest value?

**c** How many students scored 4 marks in the quiz?

**d** What was the most frequent value?

**e** Draw a column graph of this data.

**6** 30 customers were asked to name their favourite car colour. B = black, S = silver, W = white, U = blue, R = red. C

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B | B | W | R | W | R | B | S | U |
| W | R | B | S | B | U | W | B | W |
| S | B | B | R | R | U | W | B | B |

**a** Complete a frequency table for this data.

**b** Draw a divided bar graph for this data.

**c** Which is the most popular colour for this group of customers?

**7** A sample of women were asked how many siblings (brothers and sisters) they had. C

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 4 | 0 | 2 | 3 | 2 | 4 | 3 | 5 | 4 | 3 |
| 0 | 0 | 1 | 2 | 3 | 2 | 2 | 1 | 4 | 3 |
| 1 | 1 | 1 | 5 | 2 | 5 | 0 | 1 | 2 | 1 |
| 0 | 1 | 3 | 2 | 3 | 4 | 1 | 1 | 3 | 3 |

**a** Complete a frequency table for this data.

**b** What was the highest number of siblings?

**c** What was the mode (the most frequent value)?

**d** Illustrate this data on a sector graph.

Technology
Creating a graph

## Technology

### Graphing life expectancy

Life expectancy describes the age to which a person is expected to live, based on statistical data covering different factors.

**1** Enter the data below into a new spreadsheet. It compares the life expectancy at birth for males and females in 10 countries.

| | A | B | C | D |
|---|---|---|---|---|
| 1 | Life expectancy at birth | | | |
| 2 | | | Years | |
| 3 | Country | Male | Female | Total |
| 4 | Australia | 80.5 | 85.0 | 82.7 |
| 5 | Bangladesh | 72.0 | 76.5 | 74.2 |
| 6 | Brazil | 71.2 | 78.4 | 74.7 |
| 7 | Canada | 81.1 | 85.9 | 83.4 |
| 8 | Chad | 56.5 | 60.1 | 79.4 |
| 9 | China | 74.0 | 78.4 | 76.1 |
| 10 | India | 68.4 | 71.2 | 69.7 |
| 11 | Spain | 79.0 | 85.2 | 82.0 |
| 12 | Sweden | 80.4 | 84.5 | 82.4 |
| 13 | USA | 78.0 | 82.5 | 80.3 |

Source: www.cia.gov

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2 Highlight all 4 columns, including headings. Click **Insert** and **2-D Column Graph**. Give the graph an appropriate title and label the axes.

3 Which country has:

- **a** the lowest total life expectancy?
- **b** the highest total life expectancy?
- **c** the lowest male life expectancy?
- **d** the lowest female life expectancy?
- **e** the highest male life expectancy?
- **f** the highest female life expectancy?

4 Suggest reasons why some countries would have higher life expectancies than others.

5 In column E, calculate the difference between the male and female life expectancies for each country.

6 Generally, do males or females live longer?

7 Which country has the greatest difference between its male and female life expectancies?

8 Which country has the least difference?



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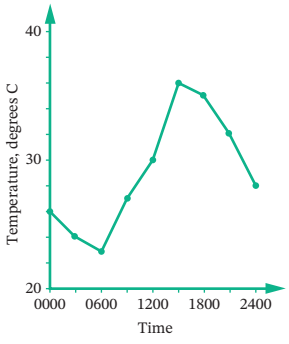


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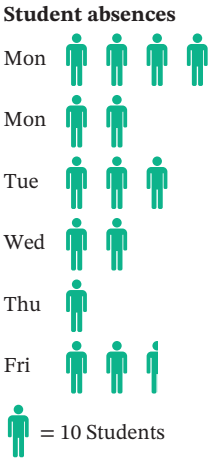


# Line graphs, pictograms and infographics 7.02

There are many different ways to display data. This table shows 3 more examples.

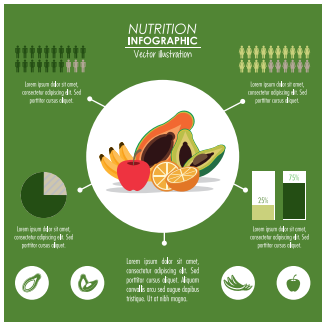
| Graph | Description |
|---|---|
| **Line graph**<br>  | • Used to show changes over time<br>• Lines connect points that describe the values of a numerical quantity<br>• The horizontal axis shows the time<br>• The vertical axis shows the quantity |

**Pictogram (or picture graph)**




- Use a picture or symbol to represent data
- Each picture represents a quantity, as shown in the key
- Mostly used for data that are in categories
- Easy to understand but not very accurate

**Infographic**




Alamy Stock Photo/Jemastock

- Uses images, graphs, and a small amount of text to illustrate data, statistics and information
- Often illustrates multiple facts
- Easy to understand

## Example 3

Display this data from Example 1 on a pictogram.

| TV sport | Australian football | Cricket | Motor racing | Rugby league | Rugby union | Football |
|---|---|---|---|---|---|---|
| Frequency | 13 | 16 | 8 | 19 | 10 | 14 |

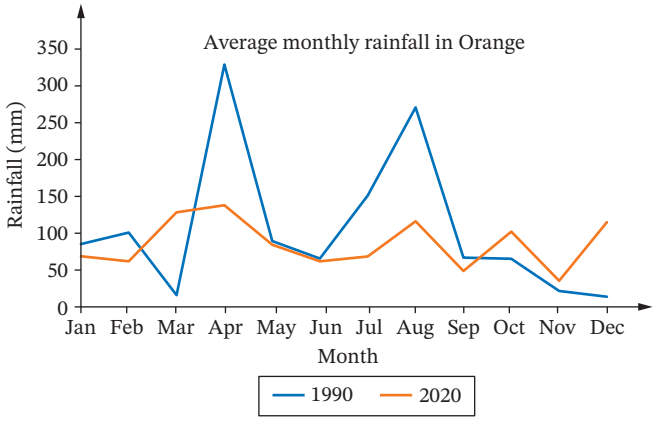
### Solution



## EXERCISE 7.02 ANSWERS ON P. 591

# Line graphs, pictograms and infographics U F R C

**1** This line graph shows the average monthly rainfall for Orange in the years 1990 and 2020. R C




Source: Bureau of Meteorology. Creative Commons License

**a** Which month in which year had the highest rainfall?

**b** Which month in which year had the sharpest drop in rainfall?

**c** In which 2 months was the rainfall approximately the same in both 1990 and 2020?

**d** In which 2 consecutive months in 1990 was the rainfall approximately the same?

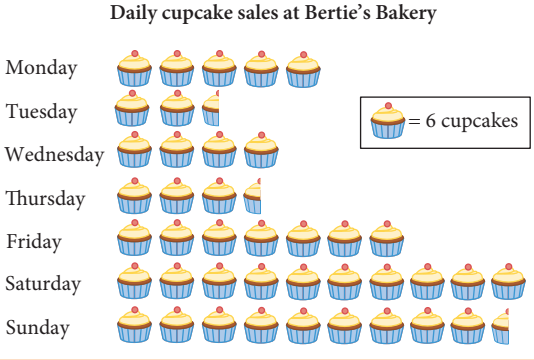
**e** Which year do you think had the higher rainfall overall: 1990 or 2020? How can you tell?

**f** You are planning a holiday in Orange. Based on these graphs, what month of the year would you choose in the hope of having least rain? Justify your answer.

**2** Draw a line graph for the data below, showing the monthly mean minimum temperatures (in °C) in Griffith in 2022. C

| Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 20 | 17 | 16 | 12 | 8 | 5 | 3 | 5 | 6 | 10 | 10 | 13 |

**3** This pictogram shows the daily sales figures for cupcakes at Bertie's Bakery for one week. R C




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**a** How many cupcakes were sold on Saturday?

**b** On which day were 42 cupcakes sold?

**c** True or false: More cupcakes were sold on the weekend than over the rest of the week combined.

**d** On which day were the least number of cupcakes sold? How many cupcakes were sold on this day?

**e** What is one disadvantage of using a pictogram to present data? Consider how you would represent 31 cupcakes being sold.

EXAMPLE 3

**4** Draw a pictogram for this data from Exercise 7.01, Question 1, about the day of the week on which students trained for their chosen sport. C

| Day | Monday | Tuesday | Wednesday | Thursday | Friday |
|---|---|---|---|---|---|
| Number of students | 3 | 13 | 4 | 7 | 3 |

**5** This is a pictogram and an infographic. C

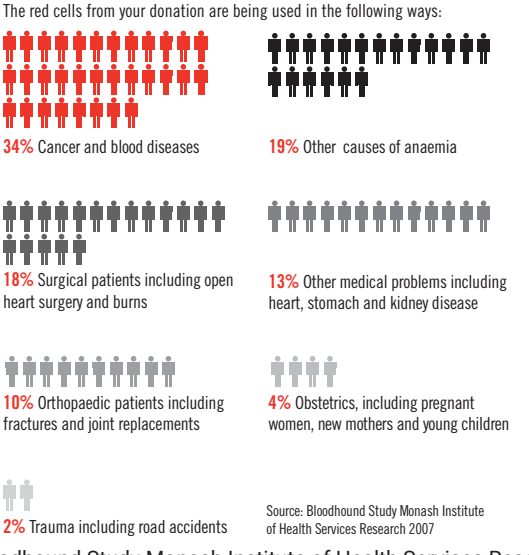



**a** What is one advantage of this type of pictogram?

**b** Design a similar pictogram and infographic to illustrate the statistic:

'In 2007, 4 in every 10 people in Australia were not connected to the internet.'

**6** This graph was published by the Australian Red Cross Blood Service. It shows where donated blood is used. R C




Source: Bloodhound Study Monash Institute of Health Services Research 2007. With permission from Australian Red Cross Blood Service.

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**a** What does each person symbol stand for?

**b** True or false?

**i** The most popular category is more than double the size of the 2nd most popular category.

**ii** The 2 most popular categories make up more than half of the use of donated blood.

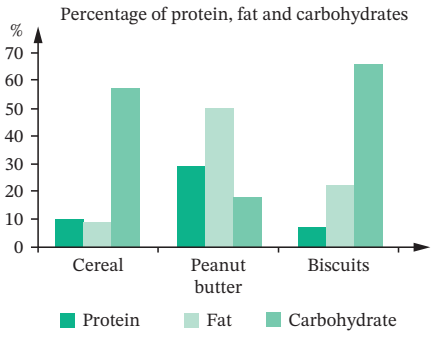
**c** What percentage of donations goes towards helping people with bone issues?

**d** Is there any information in this graph that surprises you? Explain your answer.

**e** Does this graph make you more likely to donate blood?

**f** If 27 000 L of blood are collected in one week, how much of it goes to help patients with cancer and blood diseases?

**7** We use a clustered column graph for categories we want to compare. This graph shows the nutritional information for different foods. R C




**a** Which food has the highest percentage of protein?

**b** Which food has the lowest percentage of fat?

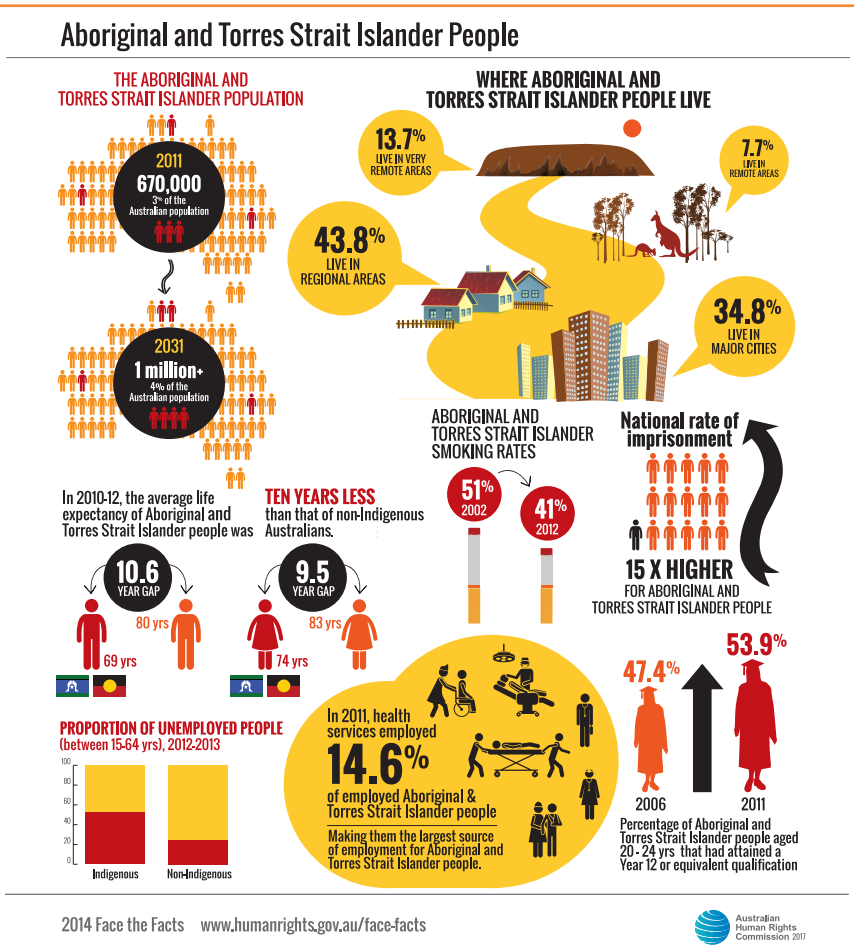
**c** If you were on a low carbohydrate diet, which food could you include in your diet?

**d** If you were on a low protein diet, which foods could be included in your diet?

**e** Jack had a 140 g serving of cereal for breakfast. Approximately how many grams of protein did the serving of cereal contain?

**f** Lizzie has to limit her carbohydrate intake. What foods should she avoid?

**8** This is an infographic about Aboriginal and Torres Strait Islander people published by the Australian Human Rights Commission. Read through it and notice the way it presents information in a clear and colourful way. R C




© Australian Human Rights Commission 2017 CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/legalcode)

**a** Which piece of information did you read first? Why?

**b** List 3 facts that you have learned from this infographic.

**c** What type of message is this infographic trying to convey?

**d** Where do most Aboriginal and Torres Strait Islander people live?

**e** What was the life expectancy of a non-Indigenous woman in 2010–12?

**f** There's a graphic that shows the population of Aboriginal and Torres Strait Islander people in the year 2031, in the future. How is it possible to have that data?

**g** Which industry was in 2011 the most popular place for Aboriginal and Torres Strait Islander people to work in?

**h** How does this infographic display data in an interesting way, as opposed to just text or just graphs?

**i** What are the features of a good infographic?

**j** Design an infographic about the students who make up your maths class.

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**9** Here is some data about Nusentry High School.

| | | | |
|---|---|---|---|
| Male students | 484 | Female teachers | 25 |
| Female students | 492 | Total teachers | 45 |
| Total students | 976 | Year 7 students | 203 |
| Students of First Nations background | 230 | Year 8 students | 194 |
| Students of Asian background | 310 | Year 9 students | 190 |
| Students who live in town | 780 | Year 10 students | 182 |
| Students who live out of town | 196 | Year 11 students | 111 |
| Male teachers | 20 | Year 12 students | 96 |

Create an infographic describing some of this data. R C

**10** According to the 2021 Australian Census, there were 167 Aboriginal and Torres Strait Islander languages spoken at home by 76 978 people. The most widely used language groups are listed below.

| Language group | Percentage of all ATSI languages |
|---|---|
| Arnhem Land and Daly River Region | 14.5 |
| Western Desert | 10.9 |
| Torres Strait Island | 12.0 |
| Yolngu Matha | 8.5 |
| Arandic | 7.4 |

Source: Australian Bureau of Statistics

Aboriginal and Torres Strait Islander people: Census 2021. CC BY 4.0 License (https://creativecommons.org/licenses/by/4.0/)

Create an infographic using this data. R C

**11** Infographics are used to convey many different types of information. Search the Internet for different types of infographics. There are also different programs which can help you make an infographic. Find the names of these.



# Types of data 7.03

Statistical data may be in the form of categories (such as hair colour or favourite sport), or in the form of numbers (such as soccer score or number of cars owned). There are 2 types of data.

**Categorical data** can be grouped into categories, such as a type of seafood or the suburb where you live.

**Numerical data** (or **quantitative data**) can be counted or measured, such as number of children or a person's height.

Puzzle
Data match-up

Foundation Standard Complex

Video
Types of data

## Example 4

Classify each of the following types of data as categorical or numerical.

**a** make of car

**b** weekly pay

**c** weight of athlete

**d** favourite radio station

### Solution

**a** Make of car is categorical data. Involves words or categories

**b** Weekly pay is numerical data. Involves numbers

**c** Weight of athlete is numerical data. Involves numbers

**d** Favourite radio station is categorical data. Involves words or categories

## Types of categorical data

Categorical data can be further divided into **ordinal data** and **nominal data**.

**Ordinal data** can be ordered in some way, for example:

school report grades – extensive, thorough, sound, basic, elementary

pain severity – no pain, moderate, severe, very severe

**Nominal data** cannot be ordered, for example, hair colour, the street where you live.

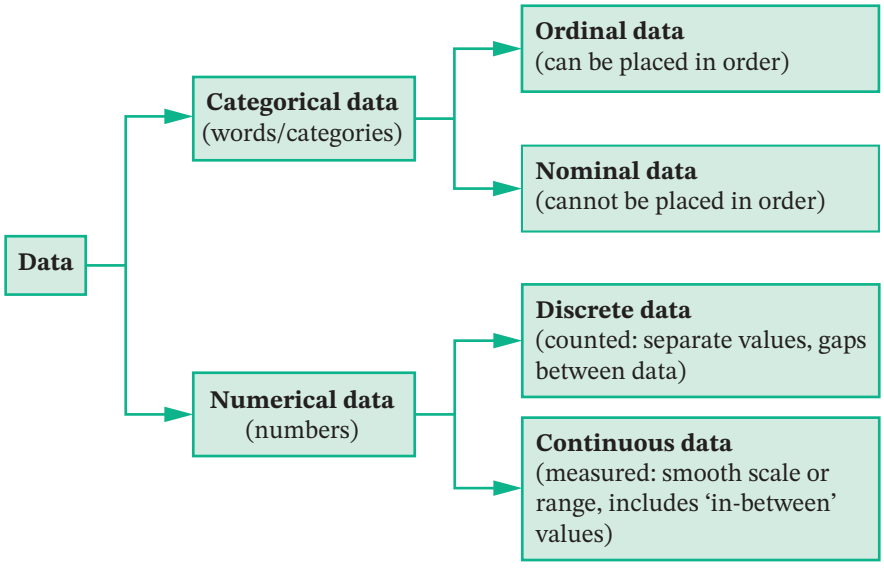
Ordinal means 'order', nominal means 'name'.

## Types of numerical data

Numerical data can be further divided into **discrete data** and **continuous data**.

**Discrete data** are counted or measured and can only take on separate, distinct values, with 'gaps' or 'jumps'. 'In-between' values are not allowed. For example, the number of children in a family is discrete because it could be 0, 1, 2, etc., but never 2.3 or $1\frac{2}{5}$.

**Continuous data** are measured on a 'smooth' scale without any gaps, and can take on a full range of values. 'In-between' values are allowed. For example, a town's temperature is continuous because it could be 28°C, 29°C, 29.5°C, 29.8°C, etc.






## Example 5

Classify each type of categorical data as nominal or ordinal:

**a** make of car (for example, Ford, BMW, Toyota)

**b** colour of Olympic medal

**c** sex of newborn baby

**d** finishing position in race

### Solution

**a** Make of car is nominal data. They cannot be placed in order.

**b** Colour of Olympic medal is ordinal data. Gold, silver, bronze have an order.

**c** Sex of newborn baby is nominal data. They cannot be placed in order.

**d** Position in race is ordinal data. 1st, 2nd, 3rd, etc. have an order.

Video
Types of data

## Example 6

Classify each type of numerical data as discrete or continuous:

**a** number of rooms in a home
**b** loudness of aeroplane
**c** shoe size
**d** weight of whale



Shutterstock.com/Paul S. Wolf

### Solution

**a** Number of rooms is discrete data. — Takes on separate values
**b** Loudness of aeroplane is continuous data. — Takes on a smooth scale of values
**c** Shoe size is discrete data. — Takes on separate values
**d** Weight of whale is continuous data. — Takes on a smooth scale of values

## EXERCISE 7.03 ANSWERS ON P. 591

## Types of data U F R C

EXAMPLE 4

**1** Classify each type of data as categorical (C) or numerical (N). R C

**a** mass of student
**b** brand of computers
**c** price of ice cream
**d** number of TV sets owned by family
**e** amount of rainfall
**f** cause of tooth decay
**g** sets won in a tennis match
**h** speed of truck

EXAMPLE 5

**2** Classify each type of categorical data as ordinal (O) or nominal (N). R C

**a** size of soft drink cup (for example, large)
**b** type of road vehicle
**c** ranking of a football team on the competition ladder
**d** type of seafood
**e** political party
**f** level of agreement with an opinion (for example, strongly disagree)

EXAMPLE 6

**3** Classify each type of numerical data as discrete (D) or continuous (C). R C

**a** number of homes with solar power
**b** patients' body temperature
**c** sizes of the crowd at a rock concert
**d** star ratings of hotel
**e** times taken for athlete to run 100 m
**f** school population
**g** amount of cough medicine taken by child
**h** number of road accidents in Grafton

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**4** Which of the following is **not** discrete data? Select the correct answer **A**, **B**, **C** or **D**. C

**A** The number of shoes you own

**B** The number of blue-eyed students at your school

**C** The number of options in a multiple-choice test

**D** The maximum daily temperature

**5** Classify each type of data as categorical ordinal (CO), categorical nominal (CN), numerical discrete (ND) or numerical continuous (NC). R C

**a** hair colour

**b** customer rating of a restaurant's level of service (poor, satisfactory, good, very good)

**c** number of points a player scored in a netball match

**d** film classification (G, PG, M, MA15+, R)

**e** the age you turn this year

**f** height of bridge

**g** number of staff employed

**h** country where cars were manufactured



# Census vs sample

7.04

In statistics, the **population** means all of the items being studied, whether they be people, cars, businesses or houses. Statistical data is collected in a variety of ways, such as surveys, Internet polls, interviews and questionnaires. To collect information, statisticians usually survey a large representative group called a **sample**. For example, to investigate opinions about school uniform, we can survey a selection of students, teachers and parents. The bigger the sample, the more representative and accurate it will be.

To collect information about a whole population, **all** people or items must be surveyed. This is called taking a **census**. For example, to plan the locations of childcare centres, the government may need to count the number of babies in different areas.

**Videos**
Counting crowds
Can fish oil make you smarter?
Can you trust your IQ?
The wrong guy won

| Sample | Census |
|---|---|
| Surveys a selected representative group of people or items. | Surveys **all** people or items (in the population). |
| Only gives **approximate** information about the population. Helps us make predictions about the population. | Gives **exact** information about the population. |
| Simple and inexpensive to survey. | Complex and expensive to survey. |
| Can be done quickly. | Takes a lot of time to collect and process the information. |

In Australia, a national census is conducted every 5 years to count the Australian population and obtain important information about every person. It takes place in a year ending in 1 or 6, such as 2021 and 2026.

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## Example 7

Should a sample or a census be used to investigate each fact?

**a** The most popular car colour.

**b** Whether energy-saving light globes really use less electricity.

**c** The number of retired people living on the Gold Coast.

**d** People's views on which nation will win the Rugby World Cup.

### Solution

**a** Sample, because it is easier and faster than asking all car-owners in Australia.

**b** Sample, because it is easier and faster than testing every light globe.

**c** Census, because the exact number is required.

**d** Sample, because it is very difficult to ask all people

# Random sampling and bias

Remember that in probability, a **random experiment** is a situation where every possible outcome has the same chance of occurring. When taking a sample, it is important that each person or item in the population has an equal chance of being chosen to be surveyed. This is called **random sampling**. To find a sample of students to survey about school uniform, we could place students' names in a box, mix them up, and then choose 100 names without looking. Alternatively, we could use a computer or calculator to generate random numbers to select students.

A sample that is not random is called a **biased sample**, and is not truly representative of the population. For example, if you survey the first 100 students to arrive at school today, or the first 100 students on the school roll, then not all students in the school have an equal chance of being selected. Your sample would not represent students who arrive at school later or whose names begin with letters at the end of the alphabet.

## Example 8

A random sample of students is to be selected for their views on a new school uniform. Will each of the following sampling methods provide a sample that is representative or biased? If biased, explain why.

**a** Randomly selecting one student from each class.

**b** Selecting 20 students from your mathematics class.

**c** Selecting the students from the school's netball teams.

**d** Randomly selecting students from the school roll by rolling a die and counting down the roll.

### Solution

**a** Representative. Every student in the school has an equal chance.

**b** Biased, because your mathematics class does not represent the other Year groups (or even the other Year 8 classes) at your school.

**c** Biased, because the netball teams do not represent the whole school and will have mainly female students.

**d** Representative. Every student in the school has an equal chance.

**EXERCISE 7.04** ANSWERS ON P. 592

## Census vs sample U F R C

EXAMPLE 7

**1** Is a sample or a census more appropriate for each situation? R C

- **a** Testing coffee for taste.
- **b** Finding the number of migrants from New Zealand.
- **c** Investigating people's views on road safety.
- **d** Finding out people's opinions on whether Australia should become a republic.
- **e** Testing a flu vaccine for effectiveness.
- **f** Counting the number of Spanish-speaking children in Australian schools.
- **g** Finding the population of Gundagai.
- **h** Finding the number of people in Australia over 40, to plan for health care.

**2** For each situation, state whether a sample or a census would be more appropriate and how the data could be collected. R C

- **a** A car dealer wants to find the level of customer satisfaction with his business.
- **b** To find the most popular TV program on Friday nights.
- **c** To find the busiest airport in Australia.
- **d** A principal wishes to find the least-liked subject in the school.
- **e** The number of people who support whaling.
- **f** The average height of students in your school.
- **g** The amount of recyclable garbage each household produces in a week.
- **h** The number of times the students in your class visit a restaurant in a month.

EXAMPLE 8

**3** The parent association for Nelson Vale High School want to know what people think of a new design for the school uniform. Which of the following samples are NOT biased? Select the correct answer **A**, **B**, **C** or **D**. R

- **A** Students who visit the school office during the day
- **B** Parents as they are picking up their children after school
- **C** Randomly-selected families from an alphabetical list of school families
- **D** Families watching the school sporting teams

**4** For each survey, describe the total population, give a suitable way of choosing a representative sample, write any problems that may be encountered and explain how bias could be minimised. R C

- **a** The attitudes of people to smoking in public.
- **b** Whether girls are better than boys at mathematics.
- **c** The choice of music at a school disco.
- **d** The political parties people will vote for at the next state election.

**5** In each survey, state whether the sample is random. If not, explain why you think it is biased. R C

**a** To find participants for a survey of students' study habits, a teacher wrote a message in the daily notices asking for volunteers.

**b** To ensure that items being made in a factory were of good quality, 2 items coming off the assembly line were tested at a randomly selected time each hour.

**c** In a poll to find voters' preferences in the next election, every 100th person listed in the White pages phone database was called. If there was no answer, the next person listed was called, and so on.

**d** In a parent survey about school uniform, the school council posted letters to all parents of students at the school asking for their opinion on the issue.

**e** In a survey about the best song of the decade, listeners of a rock radio station were asked to ring and nominate a song.

**6** The student council wants to survey students on their attitude to mobile phone usage at school. 15 students from each Year group were selected to take part in the survey. R C

**a** Do you believe the results of this survey would be reliable? Explain your answer.

**b** What factors should have been considered in selecting the sample?

**c** How would you have selected the students to be surveyed?

## Technology

### Random numbers

Random numbers can be used to select people or items for a sample. A calculator or spreadsheet can be set up to generate a random number from 1 to 100, for example, and that number can be matched to the name of a person or item from a list. For a scientific calculator:

| Casio | Sharp |
|---|---|
| Enter the formula RanInt#(1,100) as shown: ALPHA . ( 1 SHIFT ) 100 ) = | 2nd F 7 (RANDOM), select 3 R-INT, + 1 = |

This generates a random number from 1 to 100. Press = more times to generate more random numbers.

Alternatively, on a **spreadsheet,** type **=INT(RAND()*100+1)** into a cell and press the ENTER key, then the F9 key repeatedly for more random numbers.

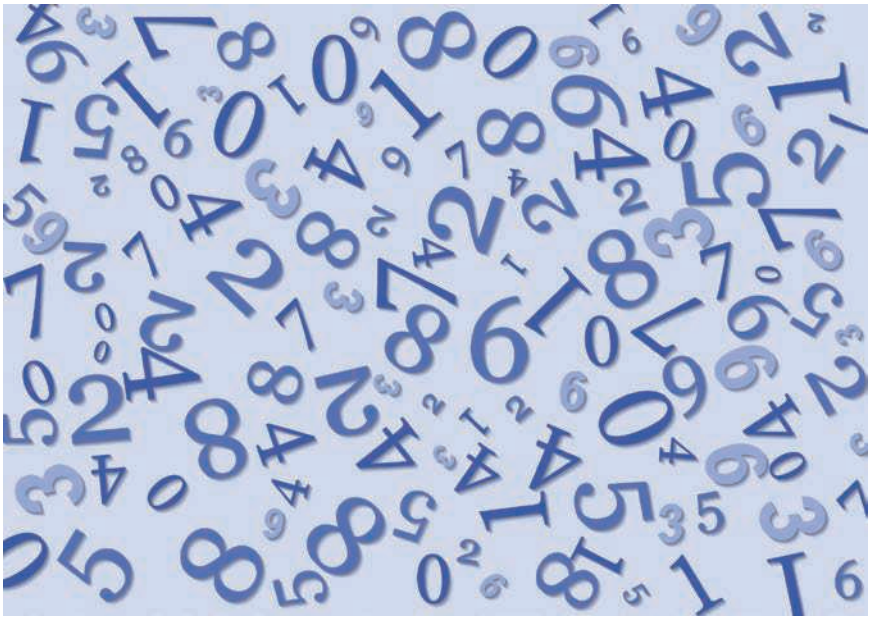
**1** Use your calculator or spreadsheet to generate 12 random numbers from 1 to 100.

**2** Use your calculator or spreadsheet to generate 12 random numbers from 1 to the number of students in your class.

**3** Use a class roll to number the students in your class from 1 onwards. Suppose you need to choose a random sample of 9 students from your class for a survey on homework. Use your calculator or spreadsheet to generate 9 random numbers to select the 9 students for your survey.

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**4** Choose 4 students from each Year 8 maths class to answer some questions.
Devise several different methods to ensure that your selection is random. Prepare a short class talk about your methods.



iStock.com/Andrii Zakharov

## Investigation

### Statistics in the media

We live in a world of 24-hour news: from TV, newspapers, websites, Facebook, Twitter, blogs, etc. To detect bias in news, we need to consider where the news item comes from and what types of samples the statistics are based on. Is the information supplied by a reporter, the police, a company executive, a government official or an opinionated blogger? Are the statistics based on a small sample, large sample, unrepresentative sample, or a phone or Internet poll of volunteers? Each may have a particular bias that may influence how the story is reported.

Find examples of news items or surveys reported in a newspaper, magazine or the Internet and investigate the following questions.

**a** Where did it come from?

**b** Who wrote the story?

**c** Does it show any bias?

**d** Can it be supported by other news providers?

**e** What type of sampling was used? Was it representative?

**f** How many people were questioned?

**g** When was the survey conducted?




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## Did you know?

### The Australian Census

The first population counts in Australia were known as musters, where people assembled at a location to be counted. The first census when people were recorded at their dwellings was in NSW in 1828. The first simultaneous census across all states was in 1881 and the population was counted as 2.25 million.

Following Federation in 1901, the first national census was developed and undertaken by the Commonwealth in 1911. From 1961, the Census has been held every 5 years in the years ending in 1 and 6.

The current census asks questions relating to the family, where you live, income, religion, level of education and other aspects of modern life.



Newpix.com/Andy Tyndall

**Find 5 questions that were asked in the most recent census.**

# 7.05 The mean and mode

**Skillsheet**
Statistical measures

**Puzzle**
Looking for gold

**Video**
Most popular pet

The mean, mode and median are known as **measures of centre** (averages) or **measures of central tendency**: they are typical or central values that represent all of the data in the set.

The range is a **measure of spread**: it describes how spread out the values are.

The **mean** is found by adding all the data values and dividing by the number of values.

## The mean

The symbol for the mean is $\bar{x}$.

$$\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$$

- The mean is a single value that represents all of the data.
- *All* of the data values are used in the calculation of the mean.
- The mean is *not* necessarily one of the data values.

The **mode** is the most common or frequent value (or values). A set of data may have more than one mode, or no mode at all.

## The mode

The **mode** is the data value (or values) that occurs the most often: the value with highest frequency.

*Think:* mode = '**m**ost **o**ften'

- The mode can be one or more values.
- The mode does not depend on all of the values in the data set.
- The mode is one (or more) of the data values.

## Example 9

The number of goals scored by Courtney's soccer team in each match over a season were:

3 2 1 4 0 2 3 1 3 1 1 0

**a** Find the mean. **b** Find the mode.

### Solution

**a** Mean $\bar{x} = \frac{\text{sum of data values}}{\text{number of values}}$

$= \frac{3+2+1+4+0+2+3+1+3+1+1+0}{12}$

$= \frac{21}{12}$

$= 1.75$

Note that the value of the mean is at the centre of the data.

**b** The mode is 1. It is the most common value (occurs 4 times).

# The mean from a calculator's statistics mode

Scientific and graphics calculators have a statistics mode (SD or STAT). Follow the instructions in the table below to calculate the mean of the values from Example **5** using your calculator's statistics mode.

| Operation | Casio scientific | Sharp scientific |
|---|---|---|
| Start statistics mode. | MODE STAT 1-VAR | MODE STAT = |
| Clear the statistical memory. | SHIFT 1 Edit, Del-A | 2nd F DEL |
| Enter data. | SHIFT 1 Data to get table 3 = 2 =, etc. to enter in column. Press AC to leave table. | 3 M+ 2 M+, etc. |
| Calculate the mean. ($\bar{x} = 1.75$) | SHIFT 1 Var $\bar{x}$ = | RCL $\bar{x}$ |
| Check the number of values. ($n$ = 12) | SHIFT 1 Var $n$ = | RCL $n$ |
| Return to normal (COMP) mode. | MODE COMP | MODE 0 |

## EXERCISE 7.05 ANSWERS ON P. 592

## The mean and mode U F R C

EXAMPLE 9

**1** The residents in a street were surveyed about the number of children living in each household. The results were:

2 2 1 2 0 3 2 1 1 4 1 0

**a** Find the mean, correct to one decimal place. **b** Find the mode.

**2** For each set of data, find:

**i** the mean (rounded to 2 decimal places, if needed) **ii** the mode(s).

**a** 6 8 7 6

**b** 7 9 5 2 2 6 2

**c** 11 7 7 6 1 2 4 1

**d** 96 53 96 45 64 76 54 54 96

**e** 47.31 47.20 47.24 47.21 47.28 47.26

**3** Use your calculator's statistics mode to calculate the mean of each data set in question **2**.

**4** Find the mode of each set of categorical data.

**a** Holden, Ford, Nissan, Holden, Hyundai, Ford, Hyundai, Nissan, Hyundai, Toyota, Toyota

**b** heart, spade, spade, diamond, spade, club, heart, heart, diamond, spade

**c** Chinese, Korean, Japanese, Korean, Chinese, Japanese, Japanese, Korean, Korean

**5** Why is it not possible to find the **mean** of each data set in question **4**? R C

**6** In a diving competition, the judges awarded the following scores.

6.2 7.0 8.2 8.8 7.0 9.1 7.0 8.1 8.0

**a** Find the mean of these scores, correct to 2 decimal places.

**b** Find the mode of these scores.

**c** Which measure of centre represents this set of scores better? Give a reason for your answer. R C

**7** What is the difference between the mean and the mode of the following set of scores?

20 60 20 50 40

Select the correct answer **A**, **B**, **C** or **D**.

**A** 20 **B** 12 **C** 18 **D** 2

**8** The ages of the members of the Carrozza family are:

1 3 11 25 26 30 31 52 58

The ages of the members of the Binns family are: R C

5 8 16 21 25 29 40 63 93

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**a** Calculate the mean age (correct to one decimal place) of each family.

**b** What is the main difference between these 2 sets of ages?

**c** What effect does this difference have on the means?

**9** 2 Year 8 classes scored the following marks for their PE assignments. R C

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **8Y:** | 5 | 5 | 5 | 5 | 6 | 6 | 6 | 7 | 7 | 7 | 8 |
| **8Z:** | 3 | 3 | 4 | 5 | 5 | 6 | 6 | 7 | 9 | 9 | 10 |

**a** Calculate the mean mark (correct to one decimal place) for each class.

**b** Describe the differences between the way each class has its marks spread out.

**c** Class 8X scored the following marks, but Hannah was away.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **8X:** | 4 | 5 | 5 | 6 | 6 | 6 | 7 | 7 | 7 | 10 |

What mark would Hannah have to score so that her class has the same mean as the other 2 classes?

**10** The mean points score per game by a basketball team over a season of 30 games was 98. What was the total number of points scored over the season? R

**11** The takings at the school canteen were: R

Monday $1600 Tuesday $2400 Wednesday $1500 Thursday $1750

**a** Find the average takings for these 4 days.

**b** How much should the canteen earn on Friday to make the average daily takings $2000?

**12 a** Find 4 values that have a mean of 10 and a mode of 11. R

**b** Find 7 values that have a mean of 13 and a mode of 5.



# The median and range 7.06

The **median** is the middle value of a set of data or the average of the 2 middle values.

## The median

When the values of a data set are ordered, the **median** is:

- the middle value if there is an odd number of values
- the average of the 2 middle values if there is an even number of values.

*Think:* 'Median' sounds like 'medium', which is halfway between small and large.

A 'median' strip is the strip of grass or concrete down the *middle* of a highway.

- The median is a single value that represents all of the data.
- The median does not depend on all of the data values, so it is not affected by outliers (extremely high or low values).
- The median is often one of the data values.

The **range** is a measure of the spread of a data set, and is simply the difference between the highest and lowest values.

**Skillsheet** Statistical measures

**Worksheet** Mean, median, mode

**Puzzles** Ranges and averages

Data puzzles

Mode, median and mean

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**Videos**
The mean, mode, median and range

The mode, median and mean

Average Joe

**Interactive**
Centre and variability

## The range

Range = highest value − lowest value

## Example 10

The ages of the members of the Carrozza family are:

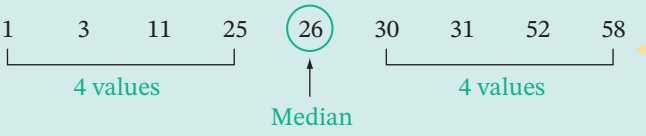
3 1 25 58 26 52 30 31 11

Find:

**a** the median **b** the range

### Solution

**a** First, rewrite the ages in order.




Note that the value of the median is at the centre of the data.

There are 9 values, an odd number, so there is one middle value.

$\frac{1}{2} \times 9 = 4\frac{1}{2}$, so round up to 5 and count to the 5th value (circled above) to find the median.

Median = 26

**b** Range = 58 − 1
= 57

Highest value − lowest value

## Example 11

The lap times (in seconds) of 6 cyclists were:

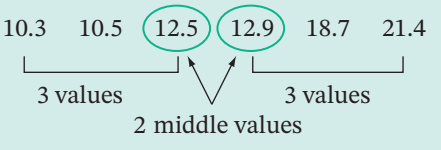
12.5 21.4 10.5 18.7 12.9 10.3

Find:

**a** the median **b** the range

### Solution

**a** Rewrite the values in order:




There are 6 values, an even number, so there are 2 middle values.

$\frac{1}{2} \times 6 = 3$, so count to the 3rd and 4th values (circled above) and find their average.

$$\text{Median} = \frac{12.5 + 12.9}{2}$$
$$= 12.7$$

12.7 is halfway between 12.5 and 12.9

**b** $\text{Range} = 21.4 - 10.3$
$= 11.1$

Highest value − lowest value



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7.06

## EXERCISE 7.06 ANSWERS ON P. 592

### The median and range U F PS R C

**1** What is the median of this set of data?

10 3 5 9 4 3 10

Select the correct answer **A**, **B**, **C** or **D**.

**A** 3 **B** 5 **C** 7 **D** 10

EXAMPLE 10

**2** What is the range of this set of data?

17.1 18.3 14.5 11.8 21.9 14.5

**A** 16.4 **B** 15.8 **C** 14.5 **D** 10.1

**3** For each set of data, find:

**i** the median **ii** the range.

**a** 11 9 7 4

**b** 46 50 48 47 50 40

**c** 12 13 11 14 10

**d** 84 50 0 28 14 99 51 66 51

**e** 57 57 57 57 57 57 57

**f** 64.2 99 99.2 71.2 93.5 61.8

EXAMPLE 11

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**4** Manish likes to gamble on horse races. His winnings or losses (in dollars) each day are as follows (losses are shown as negative values). R C

| | | | | | | |
|---|---|---|---|---|---|---|
| 8 | 21 | −16 | −25 | 21 | −25 | 16 |
| 11 | −12 | 12 | −25 | 17 | −25 | 14 |

**a** Calculate the mean, correct to the nearest cent.

**b** Find the median and the mode.

**c** As Manish wins more often than he loses, should he continue to gamble on horse races? Give a reason for your answer.

**5** 11 houses were sold in Jacaranda Avenue. The selling prices are listed here. R C

| | | | | | |
|---|---|---|---|---|---|
| $580 000 | $485 000 | $660 000 | $758 000 | $644 000 | $1 670 000 |
| $815 000 | $686 000 | $666 500 | $720 800 | $805 000 | |

**a** Find the mode.

**b** Find the median.

**c** Find the mean.

**d** Which measure of central tendency best describes this set of house prices? Give a reason for your answer.

**6** 2 ten-pin bowlers recorded the following scores over 10 games. PS R C

| | | | | | |
|---|---|---|---|---|---|
| *Aldo*: | 175 | 175 | 160 | 165 | 180 |
| | 145 | 150 | 177 | 177 | 177 |
| *Peter*: | 173 | 173 | 174 | 175 | 175 |
| | 176 | 176 | 177 | 179 | 179 |


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**a** Calculate the mean for each bowler.

**b** Calculate the range for each bowler.

**c** Who would you choose, and why, if you were asked to present an award for:

**i** the more talented bowler?

**ii** the more consistent bowler?

**d** A third player, Rob, recorded the following scores over 9 games.

| | | | | | |
|---|---|---|---|---|---|
| *Rob*: | 134 | 175 | 175 | 144 | 176 |
| | 150 | 187 | 166 | 180 | |

What score would Rob need to achieve in his tenth game to have the same mean score as Aldo?

**7** The favourite party food for a group of 3-year-old children was recorded. R

| | | | | |
|---|---|---|---|---|
| popcorn | fruit | chocolate | fruit | popcorn |
| chocolate | chocolate | fruit | fruit | fruit |

Which is the only statistical measure that can be found for this data?

**A** mean **B** median **C** mode **D** range

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**8** Samir is the coach of the Panthers athletics team. He needs to pick his best runner for the 200 m race. He has recorded the times (in seconds) for the last few races of Daniel and Cristian. R C

| | | | | | | |
|---|---|---|---|---|---|---|
| **Daniel** | 29.4 | 31 | 27.5 | 32.5 | 28 | |
| **Cristian** | 28.5 | 27 | 30.5 | 28.5 | 31.5 | 28.5 |

**a** Calculate the mean for each runner.

**b** Calculate the range for each runner.

**c** Who do you think Samir should choose as the runner for the next event? Give a reason for your answer.

**9** A group of 8 friends counted the number of letters in their surnames. The results were:

4 6 7 5 4 6 9 ☐

where ☐ represents a missing value.

Find a possible value of ☐ if the median is 6 and the range is 5. PS R

**10** Find 8 values that have a median of 8 and a range of 8.

**b** Find 7 values from 3 to 9 inclusive that have a mode of 4 and a median of 6. PS R

## Technology

### Mean, mode, median and range

**1** Enter the data below into a new spreadsheet, listing the daily maximum temperatures in Camden for one week in January.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Day | Temperature (deg C) | | | |
| 2 | Saturday | 31.5 | | Mean | |
| 3 | Sunday | 32.2 | | Mode | |
| 4 | Monday | 33 | | Median | |
| 5 | Tuesday | 36 | | Maximum | |
| 6 | Wednesday | 36.8 | | Minimum | |
| 7 | Thursday | 35.9 | | Range | |
| 8 | Friday | 26.7 | | | |

**2** Enter each formula into the given cell to calculate each statistic.

Cell E2: =**average(B2:B8)**

Cell E3:=**mode(B2:B8)**

Cell E4:=**median(B2:B8)**

Cell E5:=**max(B2:B8)**

Cell E6: =**min(B2:B8)**

Cell E7: =**E5-E6**

7.06

**Technology**
Students' marks

Finding the mean, median, mode and range

Statistical measures calculator

**Spreadsheet**
Statistical measures calculator

Foundation Standard Complex

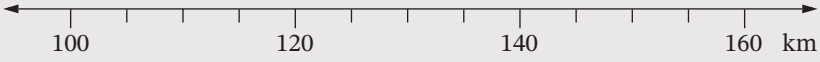
# Mental skills 7: Maths without calculators ANSWERS ON P. 592

## Reading linear scales

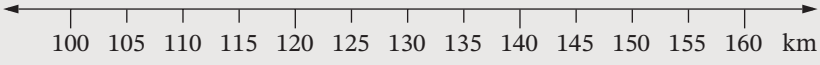
Understanding and reading the scale on a measuring instrument, on a number line or on the axis of a graph is an important mathematical skill.

**1** Study each example.

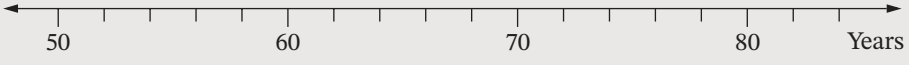
**a** Complete the missing values on this scale.



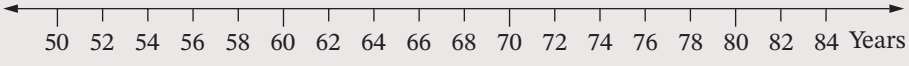

- First, choose 2 values on the scale, say 100 and 120.
- Count the number of intervals ('spaces') between the 2 values. There are 4 intervals between 100 and 120.
- To find the size of each interval, divide the difference between the 2 values by the number of intervals.
- Difference = 120 – 100 = 20 km
- Number of intervals = 4
- Size of an interval = 20 ÷ 4 = 5 km
- Use the calculated size of an interval to complete the missing values.




**b** Complete the values on this scale.



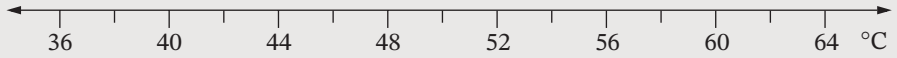

- Choose 50 and 60 on the scale.
- Number of intervals (between 50 and 60) = 5
- Difference (between 50 and 60) = 60 – 50 = 10 years
- Size of an interval = 10 ÷ 5 = 2 years.




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**2** Now copy and complete the following scales.

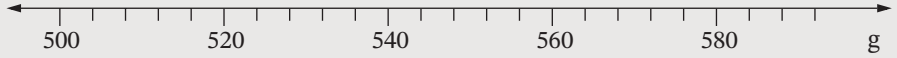
**a**



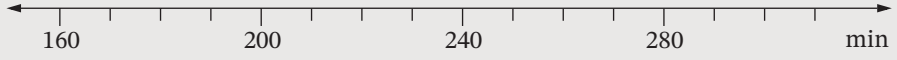

**b**



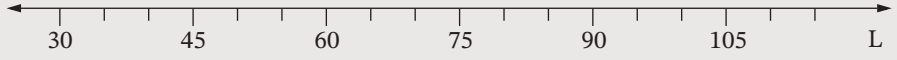

**c**



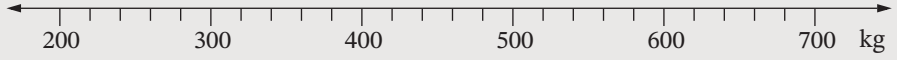

**d**



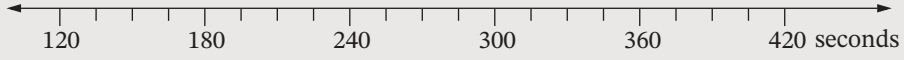

**e**




**f**




**g**




**h**




## Investigation

### Gym friends

7 friends joined a gym to lose weight.

| Name | Sex | Age | Weight loss |
|---|---|---|---|
| Brenda | F | 36 | 4 kg |
| Peter | M | 34 | 28 kg |
| Mark | M | 28 | 21 kg |
| Fiona | F | 33 | 2 kg |
| Manjeet | F | 30 | 6 kg |
| Mia | F | 34 | 3 kg |
| Natisa | F | 29 | 6 kg |



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In the following questions, round all mean answers to 2 decimal places.

**1** What is the mean age?

**2** Find the mean age of the males.

3 What is the median age of the group? Who has this age?

4 What is the median age of the males? Who has this age?

5 If the men were not counted, would the mean age or the median age be affected more?

6 Predict what would happen to the mean age if Imraan (aged 21) joined the group. Test your prediction.

7 What was the mean age 2 years ago? How does this compare to the group's mean age now?

8 Compare the mean weight loss with the median weight loss of the whole group. Who has the median weight loss?

9 Compare the mean weight loss and the median weight loss of the females only. Who has the median weight loss now?

10 When Diego joined the group, the mean weight loss of the group became 20 kg. How much weight did Diego lose?

## Investigation

### Finding the mean of a large set of data

How do we find the mean of a large amount of data? Is there a simpler way than adding all the values, then dividing the total by the number of values?

A group of students was surveyed about the number of hours they watched TV on Saturday:

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 4 | 0 | 3 | 1 | 2 | 2 | 3 | 5 | 4 | 3 | 3 |
| 4 | 1 | 0 | 6 | 6 | 3 | 4 | 3 | 1 | 5 | 4 | 3 |

We can calculate the mean of this set of data using 3 different methods.

**Method 1: Adding all the values and dividing by the number of values**

1 Find the sum of all the data values.

2 How many values are there in the data set?

3 Hence show that the mean is approximately 3.17 hours.

**Method 2: Arranging the values in order first to simplify the addition**

1 Sort all of the values in ascending order and group the equal values:

0 0 1 1 1 2 2 3 3 3 3 3 3 4 4 4 4 4 5 5 6 6 6

2 To find the sum of all the values, notice that we can multiply each value by its frequency and add those amounts together.

Evaluate $(0 \times 2) + (1 \times 3) + (2 \times 2) + (3 \times 7) + (4 \times 5) + (5 \times 2) + (6 \times 3)$.

3 Hence show that the mean is approximately 3.17 hours.

**Method 3: Using a frequency table with an extra column**

| Score | Frequency | Frequency × Score |
|---|---|---|
| 0 | 2 | 0 |
| 1 | 3 | 3 |
| 2 | 2 | 4 |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| **Total** | | |

This is a more organised version of Method **2**.

**1** Copy and complete the frequency table.

**2** What is the total number of values?

**3** What is the sum of all the values?

**4** Hence show that the mean is approximately 3.17 hours.



# Analysing frequency tables 7.07

## Example 12

The number of minutes each train was late at a railway station in one afternoon were:

2 4 1 0 0 3 2 3
3 2 0 3 2 1 3

Copy and complete this frequency table for the data and use it to find:

**a** the range
**b** the mode
**c** the median
**d** the mean, correct to one decimal place.

| Score, $x$ | Frequency, $f$ | $fx$ |
|---|---|---|
| 0 | | |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| Total | | |

**Worksheet**
The mean and *fx* tables

**Videos**
Analysing frequency tables

Statistics from a frequency table

## Solution

$fx$ means $f \times x$, so in the $fx$ column, multiply each score ($x$) by its frequency ($f$):

$3 \times 0 = 0$

$2 \times 1 = 2$

$4 \times 2 = 8$

| Score, $x$ | Frequency, $f$ | $fx$ |
|---|---|---|
| 0 | 3 | 0 |
| 1 | 2 | 2 |
| 2 | 4 | 8 |
| 3 | 5 | 15 |
| 4 | 1 | 4 |
| Total | 15 | 29 |

**a** Range $= 4 - 0$
$= 4$

Highest value − lowest value

**b** Mode $= 3$

The score with the highest frequency (5)

**c** There are 15 data values (an odd number), so the median (middle) is the **8th** value. According to the frequency table, the values in order are:

0, 0, 0  1, 1  2,2,(2),2  3,3,3,3,3  4

Half of 15 is $7\frac{1}{2}$, so round up to 8 and count the 8th value.

The first 3 values are 0, the next 2 values are 1, the next 4 values are 2, and so on.

$\therefore$ The 8th value (circled) is 2.

Median $= 2$.

**d** Mean $\bar{x} = \dfrac{\text{sum of values}}{\text{number of values}}$.

The $fx$ column multiplies each score by its frequency, so its total, 29, is the sum of all the data values. If we write all the values, we can see that the $fx$ column groups and adds the values.

$\underbrace{0,0,0}_{\text{Sum } 0}\ \underbrace{1,1}_{2}\ \underbrace{2,2,2,2}_{8}\ \underbrace{3,3,3,3,3}_{15}\ \underbrace{4}_{4}$

The $f$ column shows the frequency of each value so its total, 15, is the number of values.

$$\text{Mean} = \frac{\text{sum of } fx}{\text{sum of } f}$$
$$= \frac{29}{15}$$
$$= 1.9333\ldots$$
$$= 1.9$$

## Statistics for data in a frequency table

- the **mean** is $\bar{x} = \dfrac{\text{sum of } fx}{\text{sum of } f}$
- the **mode** is the data value(s) with the highest frequency, $f$
- the **range** is the difference between the last and first values in the score ($x$) column

## Example 13

The ages of children at a play centre are shown in the frequency table. Copy and complete the table and use it to find:

| Score, $x$ | Frequency, $f$ | $fx$ |
|---|---|---|
| 5 | 3 | |
| 6 | 4 | |
| 7 | 5 | |
| 8 | 2 | |
| 9 | 2 | |
| Total | | |



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**a** the mode

**b** the mean

**c** the median

**d** the range.

### Solution

| Score, $x$ | Frequency, $f$ | $fx$ |
|---|---|---|
| 5 | 3 | 15 |
| 6 | 4 | 24 |
| 7 | 5 | 35 |
| 8 | 2 | 16 |
| 9 | 2 | 8 |
| Total | 16 | 108 |

**a** Mode = 7

Has the highest frequency, 5.

**b** Mean:

$$\bar{x} = \frac{\text{sum of } fx}{\text{sum of } f} = \frac{108}{16} = 6.75$$

**c** There are 16 values (an even number), so there are 2 middle values.

5 5 5 6 6 6 6 (7) (7) 7 7 7 8 8 9 9

The 2 middle values

Half of 16 is 8, so count the 8th and 9th values (circled) and find the average of them.

$$\text{Median} = \frac{7+7}{2} = 7$$

Note that the mean and median are close to each other and at the centre of the values.

**d** Range = 9 − 5 = 4

# The mean of data in a frequency table using a calculator's statistics mode

Data from a frequency table can be entered into a scientific or graphics calculator using the statistics mode. Follow the instructions in the table below to calculate the mean of the children's ages from the previous page using your calculator's statistics mode.

| Operation | Casio scientific | Sharp scientific |
|---|---|---|
| Start statistics mode. | MODE STAT 1-VAR<br>SHIFT MODE scroll down to STAT Frequency? ON | MODE STAT = |
| Clear the statistical memory. | SHIFT 1 Edit, Del-A | 2nd F DEL |
| Enter data. | SHIFT 1 Data to get table<br>5 = 6 =, etc. to enter in x column<br>3 = 4 =, etc. to enter in FREQ column<br>AC to leave table | 5 2nd F STO 3 M+ 6 2nd F STO 4 M+, etc. |
| Calculate the mean. ($\overline{x} = 6.75$) | SHIFT 1 Var $\overline{x}$ = | RCL $\bar{x}$ |
| Check the number of values. ($n$ = 16) | SHIFT 1 Var $n$ = | RCL $n$ |
| Return to normal (COMP) mode. | MODE COMP | MODE 0 |

**EXERCISE 7.07** ANSWERS ON P. 593

## Analysing frequency tables U F

In this exercise, round all means to one decimal place where necessary.

EXAMPLES 12, 13

**1** This table shows how many times a sample of children visited a doctor over 3 months. Copy and complete the table, and find:

**a** the number of people surveyed

**b** the median

**c** the mean

**d** the mode

**e** the range

| No. of visits, $x$ | Frequency, $f$ | $fx$ |
|---|---|---|
| 0 | 5 | |
| 1 | 6 | |
| 2 | 14 | |
| 3 | 3 | |
| 4 | 5 | |
| 8 | 1 | |
| Total | | |

**2** A sample of hospital patients were surveyed on the number of hours they slept the previous night:

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 7 | 9 | 8 | 10 | 11 | 9 | 10 | 8 | 9 | 7 | 6 |
| 11 | 9 | 8 | 8 | 8 | 10 | 10 | 9 | 7 | 8 | 9 | 10 |

**a** How many people were surveyed?

**b** What was the highest number of hours people slept?

**c** Use a frequency table with an *fx* column to find the mean.

**d** Find the range.

**e** Find the median.

**3** The judge's scores for students' speeches (out of 10) for an English class are as shown.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 7 | 7 | 8 | 9 | 7 | 5 | 4 | 8 | 9 |
| 9 | 8 | 10 | 8 | 7 | 5 | 5 | 4 | 1 |
| 8 | 8 | 9 | 7 | 7 | 4 | 8 | 3 | 1 |

**a** Complete a frequency table for this data.

**b** What is the mode?

**c** What is the mean?

**d** What is the median?

**e** What is the range?

**4** A class of students was asked how many children there were in their families.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 5 | 1 | 2 | 3 | 1 | 3 | 2 | 4 | 2 | 4 |
| 2 | 4 | 2 | 3 | 1 | 5 | 6 | 2 | 2 | 1 | 3 | 3 |

Complete a frequency table for this data, including an *fx* column, and use it to find:

**a** the range **b** the mode **c** the median **d** the mean

**5** This frequency table shows the heights in metres of a sample of students.

| Height (m) | Frequency |
|---|---|
| 1.42 | 4 |
| 1.44 | 5 |
| 1.46 | 8 |
| 1.48 | 11 |
| 1.5 | 6 |
| 1.52 | 3 |

Find:

**a** the number of students in the sample

**b** the mean correct to 2 decimal places, using your calculator's statistics mode

**c** the range

**d** the median

**e** the mode

7.07

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**6** Carmelina recorded the number of books bought by the people entering her shop in one hour.

| No. of books | Frequency |
|---|---|
| 0 | 7 |
| 1 | 5 |
| 2 | 13 |
| 3 | 4 |
| 4 | 1 |
| 5 | 2 |
| 6 | 1 |


Shutterstock.com/SpeedKingz

What is the median number of books bought? Select the correct answer **A**, **B**, **C** or **D**.

**A** 2 **B** 3

**C** 13 **D** 4

**7** 30 people were asked how often they went to a cinema in the previous month.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 4 | 5 | 3 | 1 | 3 | 7 | 1 | 1 | 4 | 0 |
| 0 | 4 | 3 | 2 | 4 | 2 | 4 | 7 | 6 | 6 |
| 1 | 3 | 5 | 5 | 1 | 3 | 6 | 7 | 0 | 3 |

Construct a frequency table and use it to find:

**a** the range **b** the mode **c** the median **d** the mean.

**8** A pigeon breeders' club has 25 members. This frequency table shows the number of pigeons each member has.

| Score, *x* | Frequency, *f* |
|---|---|
| 8 | 3 |
| 10 | 4 |
| 14 | 3 |
| 15 | 6 |
| 17 | 2 |
| 18 | 4 |
| 20 | 3 |

ÆÆ 
iStock.com/CaoChunhai

**a** What is the range?

**b** What is the median?

**c** Use your calculator's statistics mode to find the mean.

## Technology

### Daily temperatures

1 Enter the following data into a spreadsheet: a frequency table showing the maximum daily temperatures in Coffs Harbour in July.

2 How many temperatures are in this data set? In cell B11, enter an appropriate formula for the sum of the Frequency column.

3 To calculate the mean, first complete the *fx* column. In cell C2, enter the formula for multiplying the values in the A and B columns, then **Fill Down** to copy the formula into cells C3 to C10.

4 In cell C11, enter an appropriate formula for the sum of the *fx* column.

| | A | B | C |
|---|---|---|---|
| 1 | Temperatures (x) | Frequency (f) | *fx* |
| 2 | 15 | 1 | |
| 3 | 16 | 3 | |
| 4 | 17 | 3 | |
| 5 | 18 | 7 | |
| 6 | 19 | 7 | |
| 7 | 20 | 2 | |
| 8 | 21 | 7 | |
| 9 | 22 | 0 | |
| 10 | 23 | 1 | |
| 11 | Total: | | |
| 12 | Mean: | | |
| 13 | Range: | | |

5 In cell B12, calculate the mean by entering the formula for dividing the sum of the *fx* column by the sum of the *f* column.

6 In cell B13, calculate the range by entering the formula for subtracting the lowest temperature from the highest temperature.



# Dot plots and stem-and-leaf plots 7.08

## Dot plots

A **dot plot** is a graph that uses dots to show the frequency of each data value. A dot plot shows:

- any gaps in the data
- any **clusters**, where values are grouped or bunched together
- any **outliers**: extremely high or low values that are very different from the rest of the data
- how the values are spread out.

An **outlier** 'lies outside' the other values in the data set. Outlier is pronounced 'out-ly-er'

**Worksheets**
Stem-and-leaf plots
What data?

**Presentation**
Stem-and-leaf plots

**Videos**
Analysing dot plots
Dot plots

## Example 14

An online music store recorded the daily sales figure of a particular album while it was popular, and displayed the results in a dot plot.

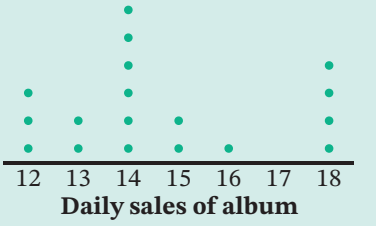
**a** On how many days were there exactly 13 sales?

**b** What is the range?

**c** Find the mode.

**d** Find the median.

**e** Calculate the mean, correct to one decimal place.



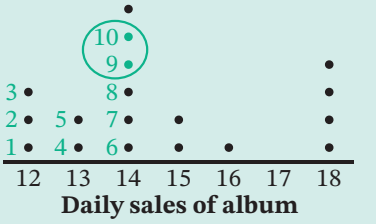

### Solution

**a** 13 sales were made on 2 days — 2 dots above 13

**b** Range = 18 − 12 = 6 — Highest value − lowest value

**c** Mode = 14 — The value with the most dots.

**d** There are 18 values (18 dots). This is an even number, so there are 2 middle values (the 9th and 10th values). By counting the dots, or by crossing out pairs of dots at each end, we can see that the middle values (circled) are 14 and 14.




$$\text{Median} = \frac{14+14}{2} = 14$$

**e** Mean: $\bar{x} = \dfrac{\text{sum of values}}{\text{number of values}}$ $\left(\text{or } \dfrac{\text{sum of } fx}{\text{sum of } f}\right)$

Number of values = 18 — 18 dots

$$\bar{x} = \frac{3\times12+2\times13+6\times14+2\times15+16+4\times18}{18}$$
$$= 14.6666\ldots$$
$$= 14.7$$

Note that the mean and the median are close to each other and at the centre of the data.

*Note*: This answer can also be found using a frequency table and/or your calculator's statistics mode.

**Videos**
Stem-and-leaf plots 1
Stem-and-leaf plots 2

## Stem-and-leaf plots

A **stem-and-leaf plot** (or **stemplot**) is like a sideways column graph, but one that lists the actual values on the horizontal columns. An **ordered** stem-and-leaf plot shows:

- all the data values, listed from smallest to largest
- any modes (the most common values)
- any clusters or outliers
- how the data are spread out.

## Example 15

The heights (in centimetres) of a sample group of Year 8 boys and girls were measured.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| *Girls*: | 148 | 155 | 153 | 137 | 155 | 137 | 146 | 165 | 157 | 168 |
| *Boys*: | 140 | 152 | 143 | 139 | 166 | 144 | 136 | 151 | 130 | 138 |

**a** Show this data on a back-to-back stem-and-leaf plot.

**b** Find the median for each set of data.

**c** Find the range for each set of data.

**d** Comment on the differences between the data for girls and boys.

### Solution

**a** Heights of Year 8 students in centimetres

| Girls | | Boys |
|---|---|---|
| 7 7 | 13 | 0 6 8 9 |
| 8 6 | 14 | 0 3 4 |
| 7 5 5 3 | 15 | 1 2 |
| 8 5 | 16 | 6 |

**b** Median: There are 10 values in each set.

The median is the average of the 5th and 6th values (circled in red on the plot).

Median: Girls $= \frac{153+155}{2} = 154$

Boys $= \frac{140+143}{2} = 141.5$

**c** Range: Girls $= 168 - 137 = 31$

Boys $= 166 - 130 = 36$

**d** The girls in the sample are generally taller (higher median), but the boys' heights are more spread out (higher range).

Video
Back-to-back stem-and-leaf plots

7.08

**EXERCISE 7.08** ANSWERS ON P. 593

## Dot plots and stem-and-leaf plots U F PS R C

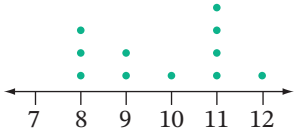
In this exercise, round all mean answers to one decimal place where necessary.

EXAMPLE 14

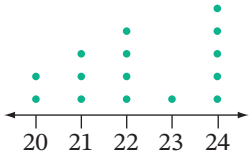
**1** For each dot plot, find:

**i** the range **ii** the mode **iii** the median **iv** the mean.

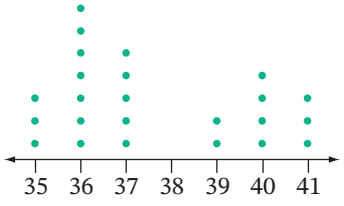
**a**



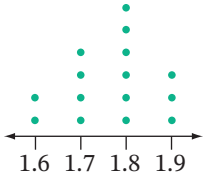

**b**




**c**




**d**




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**2** The number of goals a football player kicked for each match last season were:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 7 | 3 | 2 | 4 | 0 | 3 | 2 | 4 | 3 | 5 |
| 3 | 1 | 1 | 2 | 4 | 2 | 5 | 3 | 0 | 1 | 0 |

**a** Draw a dot plot to represent this data.

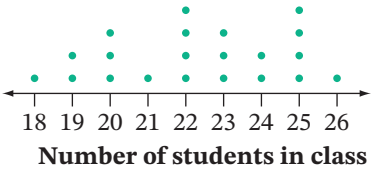
**b** How many matches were there last season?

**c** Which value is an outlier?

**d** Calculate the median.

**e** Calculate the mean.

**3** This dot plot shows the number of students in each class at Northvale Primary School.



**Number of students in class**

**a** How many students are in the smallest class?

**b** Find the range.

**c** Find the mode.

**d** Calculate the mean.

**e** What percentage (to the nearest whole number) of classes have over 20 students?

**f** Which of the following values is the median? Select the correct answer **A**, **B**, **C**, or **D**.

**A** 11 **B** 22 **C** 22.5 **D** 23

**4** The number of songs Udani downloaded each day was: R

1, 2, 0, 2, 3, 3, 3, 1, 1, 1, 6, 2, 3, 0

**a** Draw a dot plot for this data.

**b** How many days are represented by this dot plot?

**c** What is the outlier?

**d** Find the median.

**e** Calculate the mean.

**f** If this pattern continues in the following weeks, how many songs would she expect to download on average in a week?

**5** The quiz marks out of 10 for 2 Year 8 classes are shown. R C

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| *8 Hamper:* | 3 | 2 | 0 | 1 | 5 | 8 | 6 | 7 | 6 | 3 | |
| | 5 | 4 | 5 | 6 | 7 | 9 | 2 | 5 | 7 | | |
| *8 Yen:* | 7 | 6 | 3 | 7 | 8 | 1 | 9 | 4 | 6 | 7 | 2 |
| | 7 | 2 | 8 | 10 | 9 | 9 | 5 | 7 | 8 | 9 | 10 |

**a** Draw a dot plot of the marks for each class.

**b** What is the mode of the marks for class 8 Yen?

**c** What is the median of the marks for class 8 Hamper?

**d** What is the range of the marks for class 8 Yen?

**e** Draw a frequency table of the data for class 8 Hamper and use it to calculate the mean.

**f** Which do you think is the 'better' class? Give a reason for your answer.

EXAMPLE 15

7.08

**6** For each stem-and-leaf plot below, find:

**i** the range **ii** the mode **iii** the median **iv** the mean.

**a**

| Stem | Leaf |
|---|---|
| 1 | 0 2 3 |
| 2 | 1 4 4 5 6 |
| 3 | 3 3 3 7 |
| 4 | 1 2 3 5 9 |

**b**

| Stem | Leaf |
|---|---|
| 7 | 3 4 5 7 7 |
| 8 | 1 2 2 3 8 9 |
| 9 | 0 4 4 4 4 6 7 9 |

**c**

| Stem | Leaf |
|---|---|
| 10 | 0 1 1 2 |
| 11 | 7 8 8 8 9 |
| 12 | 3 6 6 7 |
| 13 | 1 1 1 1 1 8 |
| 14 | 0 |

**d**

| Stem | Leaf |
|---|---|
| 0 | 5 5 6 |
| 1 | 1 4 4 7 7 7 7 |
| 2 | 0 3 8 8 |
| 3 | |
| 4 | 9 9 9 |

**7** A survey shows that the number of students served at the school canteen daily over a 3-week period is:

105 76 97 108 114 106 124 101
112 98 95 105 117 101 92

**a** Show this information on a stem-and-leaf plot.

**b** What is the outlier?

**c** Find the mode.

**d** What was the highest value?

**e** On what stems were the values clustered?

**f** On what percentage (to one decimal place) of days were more than 100 students served?

**8** The heights in centimetres of the students in a Science class are:

156 154 158 167 164 163 155 166 159 158 137 163
171 160 164 169 157 161 160 165 145 175 167 167

**a** Make a stem-and-leaf plot for this data.

**b** On what stem are the data clustered?

**c** How many students were measured?

**d** How many students were shorter than 150 cm?

**e** Calculate the range.

**f** Find the median.

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**9** The times taken, in minutes, to drive between 2 towns along 2 different roads are shown on this back-to-back stem-and-leaf plot. PS R C

| Fuller's Ridge Rd | | Great Southern Drive |
|---|---|---|
| 7 5 3 2 | 2 | 6 8 |
| 9 8 8 5 1 1 | 3 | 0 2 4 7 8 |
| 3 3 2 0 0 | 4 | 0 1 2 6 9 9 |
| 2 | 5 | 0 1 3 7 |
| 1 0 | 6 | 2 |
| | 7 | 1 |

**a** How many trips were made along the Great Southern Drive?

**b** How many trips made along Fuller's Ridge Road took over an hour?

**c** Which road is the faster one to take to drive between the 2 towns? Explain your answer.

**d** Which statement is true for the data sets above? Select **A**, **B**, **C** or **D**.

**A** The range is the same for both data sets

**B** There is a difference of 5 minutes in the median travel times

**C** The mean travel time for Fuller's Ridge Road is 36 minutes

**D** A travel time of 71 minutes along the Great Southern Drive is an outlier

**10** This back-to-back stem-and-leaf plot shows the number of penalties that 2 football teams received for each match last season. PS R C

| Wee Waa Reds | | Narrabri Lions |
|---|---|---|
| 9 8 4 4 2 | 0 | 7 9 9 |
| 8 6 5 5 2 2 2 | 1 | 2 3 3 6 8 |
| 3 1 1 | 2 | 5 6 8 9 |
| 0 | 3 | 0 0 1 3 |

**a** How many matches did these teams play in one season?

**b** How many times did the Narrabri Lions receive fewer than 20 penalties in a match?

**c** Find the highest number of penalties received by the Wee Waa Reds in a game.

**d** Find the mode for the Narrabri Lions.

**e** Which value was an outlier for the Wee Waa Reds?

**f** Which team generally had a higher number of penalties?

**g** Which team had a greater range in their number of penalties? Give a reason for your answer.

## Investigation

### Measures of centre

**1** Create 6 values in a data set where the mean equals the median.

**2** Create 5 values in a data set where the median equals the mode.

**3** Create 4 values where the mean, the median and mode all differ.

**4** Create 5 values where the mean, median and mode are the same.

**5** 7 judges score a gymnast out of a possible 10. Their values are:

6 7 8 6 8 9 6

The scoring system requires that the lowest and highest scores are rejected and that the mean of the remaining 5 gives the final score. What is the final score?



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**6** In 4 games Dan scores 23, 11, 17 and 26 points. What score must Dan achieve in his next game so that he can average 20 on the 5 games? (Show the working or reasoning that you use.)

**7** 11 film students rate a movie out of a possible 10. Their ratings are:

7 9 9 4 6 4 8 9 7 8 5

The group cannot agree on a group rating for the movie. In one sentence, comment on each of these suggestions.

**a** Kim thinks that the rating should be 9 (the mode) because that was the rating given the most often.

**b** Vlado thinks that the rating should be 7 (the median) because it is not swayed by extremely high or extremely low ratings.

**c** Thanh thinks that the rating should be 6.9 (the mean) because it considers the ratings of all the students.

What overall rating would you give the film based on these opinions?

**8** Find 3 numbers for which the median is 15, the mean is 13 and the range is 12.

**9** Find 5 numbers for which the median is 3, the mean is 4, and the mode is 7.

# 7.09 Frequency histograms and polygons

Numerical data can be graphed on a **frequency histogram**, a special type of column graph, or a **frequency polygon**, a special type of line graph.

**Puzzles**
Statistical match-up
Highway elephants

**Worksheet**
Analysing data

## Example 16

20 families were surveyed about how many phones they owned.

Graph this data as:

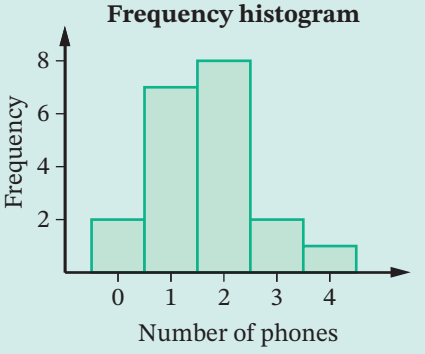
**a** a frequency histogram

**b** a frequency polygon.

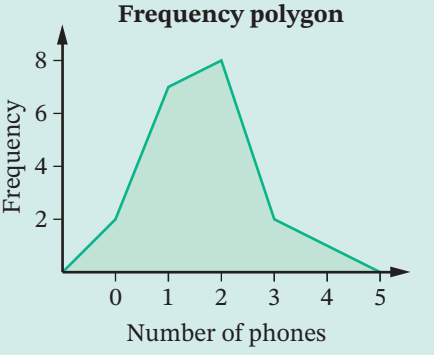
| No. of phones | Frequency |
|---|---|
| 0 | 2 |
| 1 | 7 |
| 2 | 8 |
| 3 | 2 |
| 4 | 1 |
| **Total** | 20 |

### Solution

**a**



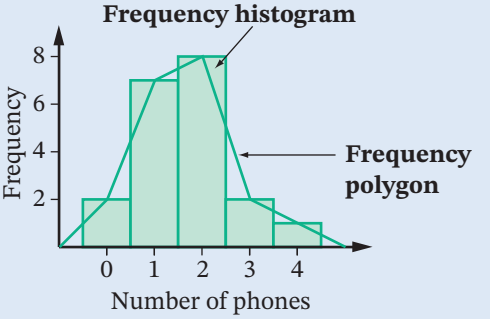

**b**




*Note:* A frequency polygon can be constructed from a frequency histogram by joining the midpoints of the tops of the columns.

## Frequency histogram

- A column graph that looks like a row of office buildings
- Frequency is shown on the vertical axis
- The values are shown on the horizontal axis
- The columns stand together without gaps between them
- The columns are centred on the values
- There is a gap of half a column width on the left (between the frequency axis and the first column)





9780170479516

## Frequency polygon

- A line graph that looks like a mountain
- Frequency is shown on the vertical axis
- The values are shown on the horizontal axis
- Begins and ends on the horizontal axis
- Called a 'polygon' because the graph and the horizontal axis together make a shape with straight sides



**EXERCISE 7.09** ANSWERS ON P. 594

## Frequency histograms and polygons **U F PS R**

EXAMPLE 16

**1** The students in a class were surveyed about how many pets they owned.

| No. of pets | Frequency |
|---|---|
| 0 | 7 |
| 1 | 11 |
| 2 | 9 |
| 3 | 3 |

**a** Graph this data as a frequency polygon.

**b** How many students were surveyed?

**c** How many students had no pets?

**d** What percentage of students had 2 pets?

**e** Find the median.

**f** Calculate the mean, correct to one decimal place.
Select the correct answer **A**, **B**, **C** and **D**.

**A** 1.0 **B** 1.3 **C** 1.5 **D** 3



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**2** Hadieya counted the number of phone calls she made each day for 30 days.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 5 | 3 | 6 | 2 | 4 | 3 | 3 | 5 | 4 | 7 |
| 5 | 1 | 6 | 3 | 2 | 3 | 5 | 6 | 1 | 6 |
| 2 | 4 | 3 | 1 | 5 | 4 | 2 | 7 | 3 | 2 |

**a** Complete a frequency table for this data.

**b** Draw a combined frequency histogram and polygon of the data.

**c** On how many days did Hadieya make 3 calls?

**d** How many phone calls were made over the 30 days?

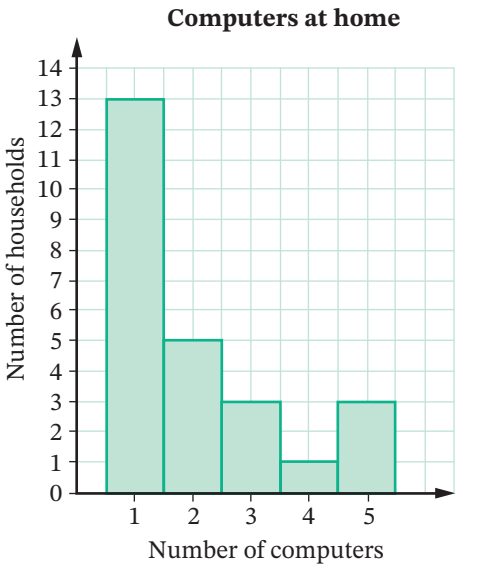
**e** On how many days did Hadieya make fewer than 3 calls?

**f** What was the mode?

**g** What was the mean (correct to one decimal place)?

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**3** This histogram shows the number of computers found in a sample of homes.




**a** How many households were surveyed?

**b** Where are most of the values clustered?

**c** Find the mode.

**d** Find the range.

**e** Complete a frequency table, including an *fx* column, for the data and use it to calculate the mean.

**f** Find the median.

**4** The numbers of goals scored by a soccer team in each match last season were:

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 0 | 1 | 2 | 3 | 4 | 3 | 1 | 2 | 0 |
| 2 | 1 | 4 | 3 | 2 | 5 | 1 | 2 | 0 | 3 |

**a** Complete a frequency table for this data.

**b** Draw a frequency histogram and frequency polygon of the data on the same axes.

**c** How many matches were played by the team?

**d** What was the highest number of goals scored?

**e** What was the mode?

**f** What was the mean?

**g** In how many matches were more than 2 goals scored?

**h** In what percentage of matches were no goals scored?



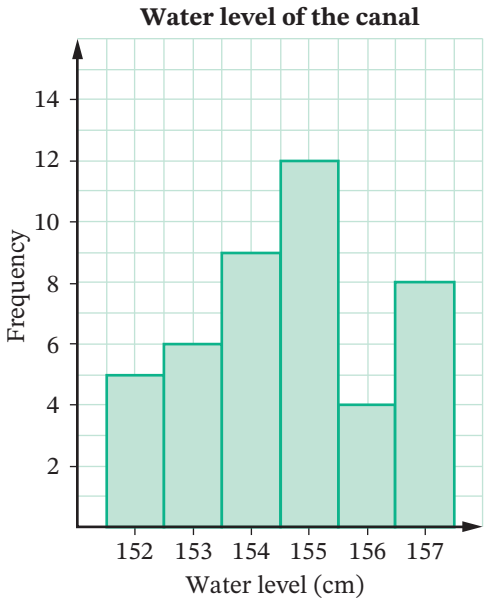
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**5** The water level of a canal was recorded over 44 days and graphed on this histogram. PS R

**a** Use the histogram to copy and complete the frequency table.

| Water level (cm) | Frequency |
|---|---|
| 152 | |
| 153 | |
| 154 | |
| 155 | |
| 156 | |
| 157 | |
| Total | |




**b** On how many days was the water level 156 cm high?

**c** What is the range?

**d** What is the mode?

**e** What is the mean, correct to 2 decimal places?

**f** On how many days was the water level lower than 154 cm?

**g** What is the median?

**h** There needs to be at least 155 cm of water in the canal for a boat to travel on it safely. On how many days would the boat be able to use the canal?

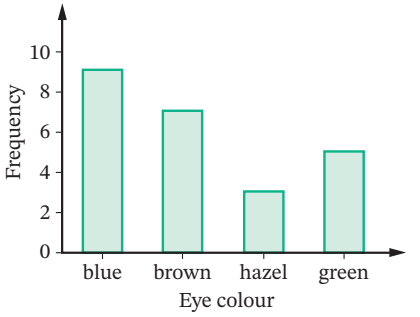
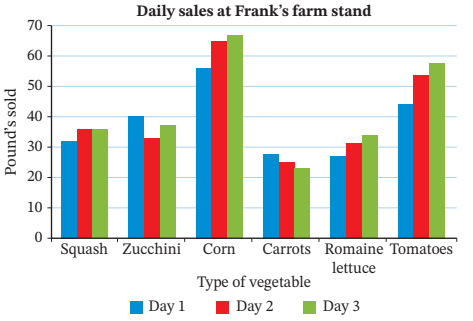
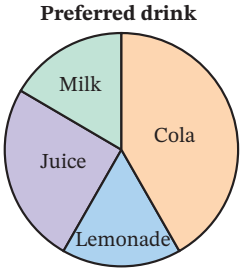
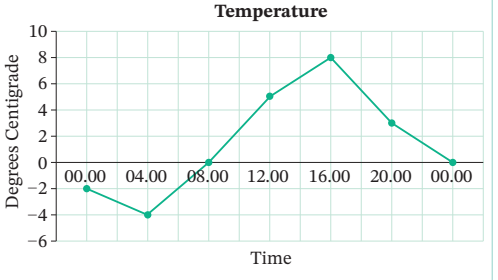
7.10

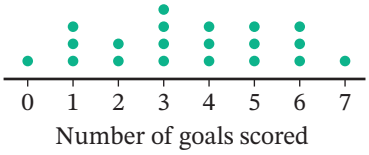
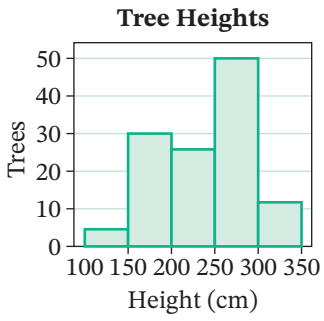
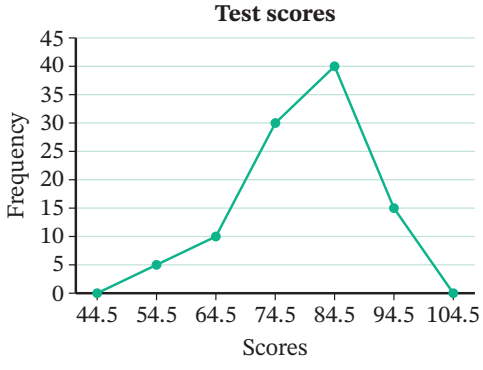
# Choosing the best graph 7.10

The best graph to use to represent a data set depends on the type of data being analysed. Here are some examples of the most appropriate graphs to use.

| Graph | Appropriate for | Description |
|---|---|---|
| **Pictogram**<br>Mode of transport: Bus ☺☺☺☺☺☺☺; Car ☺☺☺☺; Walking ☺☺☺☺☺☺; Bicycle ☺☺☺<br>**Key:** ☺ Represents 4 children | Simple data that doesn't require a lot of detail<br>Categorical data | • Uses pictures or symbols to represent frequencies<br>• Has a key to show what one picture represents<br>• Not precise |

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| Graph | Appropriate for | Description |
|---|---|---|
| **Column graph (or bar chart)**<br>  | Categorical data | • Uses columns to show frequency, gaps between columns<br>• Frequency shown on the vertical axis |
| **Clustered column graph**<br>  | Comparisons to be made between 2 or more data sets<br>Categorical data | • A key must be included to identify what the different colour bars represent |
| **Sector graph (or pie chart)**<br>  | Showing parts of a whole<br>Categorical data<br>Discrete numerical data | • Shows parts of a whole in circle form<br>• Uses sectors of a circle to show sizes of parts |
| **Divided bar graph**<br>  | Showing parts of a whole<br>Categorical data<br>Discrete numerical data | • Shows parts of a whole in rectangular form<br>• Uses sections of a rectangle to show sizes of parts |
| **Line graph**<br>  | Changes in a quantity over time<br>Numerical data | • Time on the horizontal axis, quantity on the vertical axis |

| Graph | Appropriate for | Description |
|---|---|---|
| **Dot plot**<br>  | Small data sets with low range and frequencies<br>Discrete numerical data | • A simplified column graph |
| **Stem-and-leaf plot**<br>**Stem** \| **Leaf**<br>0 \| 1 1 2 2 3 4 4 4 5 8<br>1 \| 0 0 0 1 1 3 7 9<br>2 \| 5 5 7 7 8 8 9 9<br>3 \| 0 1 1 1 2 2 2 4 5<br>4 \| 0 4 8 9<br>5 \| 1 6 7 7 8<br>6 \| 3 6<br>key: 6\|3 = 63 years old | Data sets with low frequencies<br>Discrete numerical data | • Lists every value in the data set |
| **Frequency histogram**<br>  | Large data sets with high frequencies<br>Numerical data | • No gaps between columns<br>• A space at the beginning and end of the graph<br>• Frequency is shown on the vertical axis |
| **Frequency polygon**<br>  | Large data sets with high frequencies<br>Numerical data | • Frequency is shown on the vertical axis<br>• Graph starts and finishes on the horizontal axis |

## Example 17

Identify what type of graph is appropriate to represent each data set.

**a** The number of coffees sold by a café on each day of the week

**b** The hourly temperature of a suburb over a day

### Solution

**a** Pictogram or column graph.

Days of week is categorical data and number of coffees sold will be frequency. A pictogram or column graph would easily show the differences in sales between the days.

**b** Line graph.

This is best to show a changing quantity over time.

**EXERCISE 7.10** ANSWERS ON P. 594

## Choosing the best graph U F R C

EXAMPLE 17

**1** What is the best graph to use for each data set? Justify your answer. R C

**a** Baby's length each month for the first 12 months

**b** Science test marks for all Year 8 students

**c** Minimum temperature each day for a fortnight in Coffs Harbour

**d** How students in your class travel to school

**e** Favourite sport watched on TV

**f** Heights of all students in your class

**g** Costs of manufacturing a car as a percentage of the car's price

**h** Number of households that own dogs in different streets

**i** Number of bedrooms per home in Mary Street

**2** A group of Year 8 students were surveyed on the make of car owned by their families. The results are shown in the table. R C

| Make of car | Frequency |
|---|---|
| Holden | 12 |
| Ford | 10 |
| Mitsubishi | 7 |
| Subaru | 5 |
| Toyota | 8 |
| Other | 3 |

**a** What is the best graph to use for this data?

**b** Draw the graph you have chosen.

**3** The Cairns Tourist Board wants to increase its tourist numbers in winter. It records the number of rainy days in June for a number of cities. It wants a graph that is simple, easy to read and attractive. R C

| City | Rainy days in June |
|---|---|
| Cairns | 10 |
| Melbourne | 18 |
| Adelaide | 24 |
| Sydney | 15 |
| Hobart | 17 |
| Perth | 16 |
| Brisbane | 15 |

**a** What is the best graph for this data?

**b** Draw the graph you have chosen.

**4** A survey was conducted about the colour of people's cars. The results are shown in the table below: R C

| Colour | Red | Blue | White | Green | Silver | Black |
|---|---|---|---|---|---|---|
| Frequency | 42 | 16 | 53 | 5 | 45 | 32 |

**a** What type of graph would you use to represent the results of this survey?

**b** Why is this an appropriate graph to use?

**c** Draw the graph you chose in part **a**.

**d** Is there a different type of graph that can be used to represent the same data set? Explain.

**5** The table below shows the sale of fruit (in kilograms) from a fruit shop over two days. R C

| Fruit | Monday | Tuesday |
|---|---|---|
| Apples | 10 | 15 |
| Bananas | 32.5 | 22.5 |
| Mangos | 17.5 | 25 |
| Oranges | 27.5 | 35 |
| Strawberries | 15 | 20 |

**a** What display would you use to represent the results of this survey?

**b** Why is this an appropriate display to use?

**6** This table shows the temperatures over 24 hours in Alice Springs for a November day. R C

| Time | Noon | 3 p.m. | 6 p.m. | 9 p.m. | Midnight | 3 a.m. | 6 a.m. | 9 a.m. | Noon |
|---|---|---|---|---|---|---|---|---|---|
| Temperature | 32°C | 35°C | 33°C | 29°C | 26°C | 24°C | 23°C | 29°C | 34°C |

**a** What is the best graph for this data?

**b** Draw the graph you have chosen.

# 7.11 The shape of data

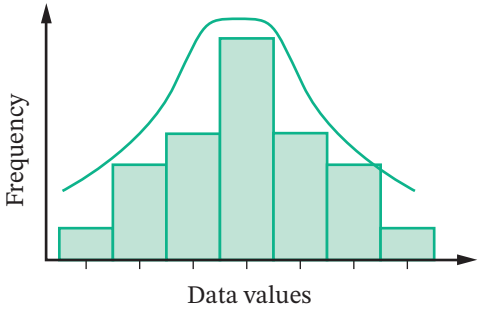
**Puzzle**
Statistical match-up

**Video**
The shape of a frequency distribution

A **statistical distribution** is the way the values of a data set are arranged, especially when graphed. When looking at histograms, dot plots and stem-and-leaf plots, an overall pattern can be seen from the shape of the display.

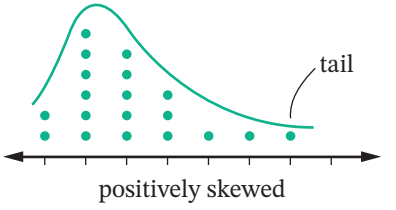
The shape of a statistical distribution shows how the data is spread and can be seen by drawing a curve around the graph or display.

A distribution is **symmetrical** if the data is evenly spread or balanced about the centre. For example, the frequency histogram and stem-and-leaf plot shown below are both symmetrical in shape. The most frequent values are the middle ones, and the least frequent values are the lower and higher ones.

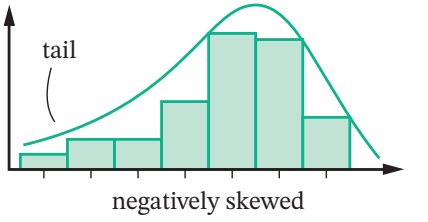



| Stem | Leaf |
|---|---|
| 2 | 6 |
| 3 | 3 7 9 |
| 4 | 5 8 |
| 5 | 0 2 3 3 7 8 |
| 6 | 0 4 |
| 7 | 6 7 8 |
| 8 | 9 |

A distribution is **skewed** if most of the data is bunched or clustered at one 'end' of the distribution and the other end has a 'tail', as shown on this dot plot and frequency histogram. The most frequent values are the lower ones for positively skewed and the higher ones for negatively skewed, and the least frequent values are at the opposite end, at the tail.




A distribution is **positively skewed** if its tail points to the right (the positive direction) and the data is bunched to the left. This could represent house prices in a country town, with mostly moderate prices and a few high prices for bigger houses and farms.

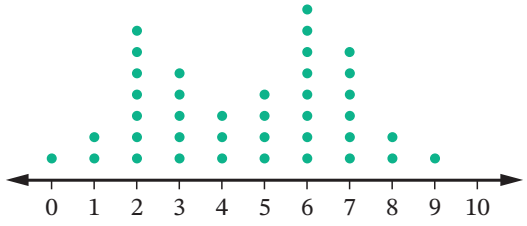



A distribution is **negatively skewed** if its tail points to the left (the negative direction) and the data is bunched to the right. This could represent the marks scored by students on an easy test, with many high marks and a few low marks.

A distribution is **unimodal** if it has one peak. All the above distributions are unimodal, and the peak indicates the mode, the data value with the highest frequency.

A distribution is **bimodal** if it has 2 peaks. The higher peak is the mode, while the other peak indicates another data value that has a high frequency.

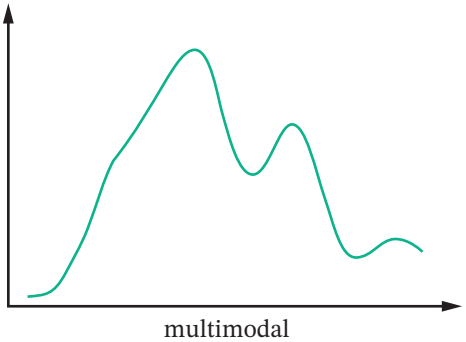
For example, this dot plot has peaks at 2 and 6 so it is bimodal. The mode, however, is 6.



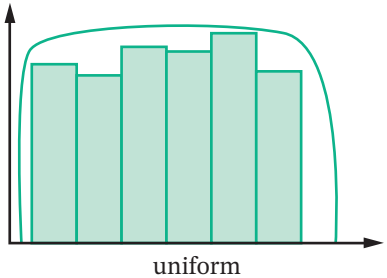

bimodal

A distribution is **multimodal** if it has 2 or more peaks (including a bimodal distribution).

A distribution is **uniform** if all the data values have roughly the same frequency. There are no peaks, and its shape is rectangular.



multimodal



uniform

The **mode** of a statistical distribution is data value that has the highest peak of the distribution.

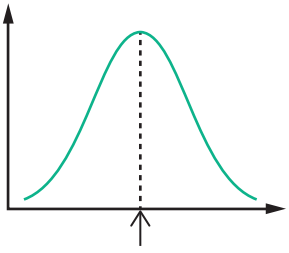
The **median** of a distribution is the data value that divides the area under the curve into half.

The **mean** of a distribution is most affected by outliers.

For a **symmetrical distribution**, the mean, median and mode are the same value.

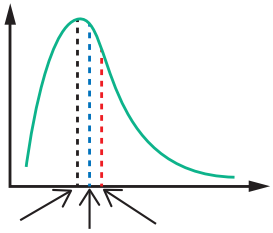
For a **skewed distribution**, the mean and median are pulled to the left or right so that the mean is further from the centre.

**symmetrical**



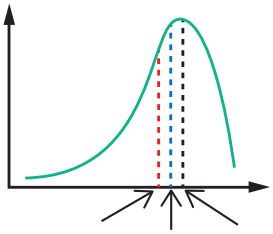
mean
median
mode

positively skewed



mode median mean

negatively skewed

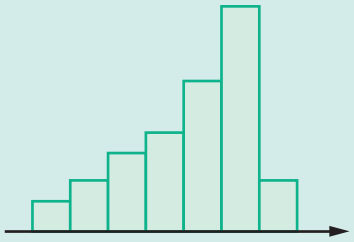


mean median mode

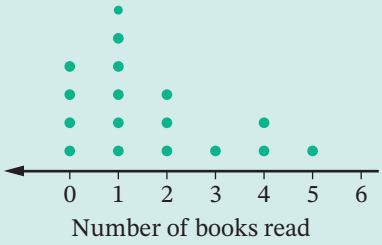
## Example 18

For each statistical distribution, describe the shape as symmetrical, positively skewed or negatively skewed.

**a**



**b**




**c**

| Stem | Leaf |
|---|---|
| 3 | 0 1 2 |
| 4 | 1 3 4 4 5 6 |
| 5 | 0 4 5 7 8 |
| 6 | 3 7 8 |
| 7 | 0 1 |
| 8 | 4 |
| 9 | 8 |

### Solution

**a** The shape is negatively skewed, tail points to the left.

**b** The shape is positively skewed, tail points to the right.

**c** The shape is positively skewed, tail points towards the higher values (down)

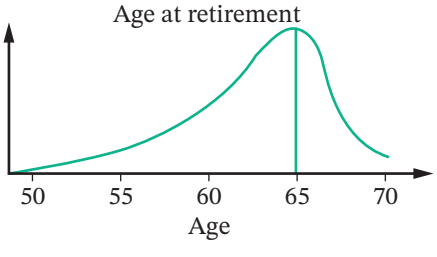
**EXERCISE 7.11** ANSWERS ON P. 595

## The shape of data UFRC

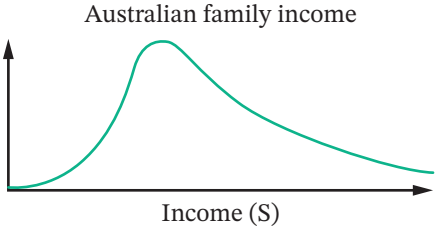
EXAMPLE 18

**1** What type of distribution is shown in each graph? Choose 2 from symmetrical, positively skewed, negatively skewed, unimodal and bimodal. **c**

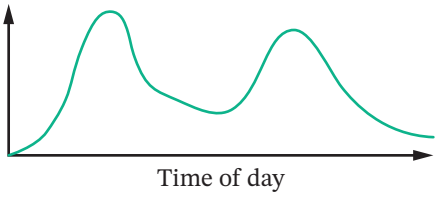
**a**



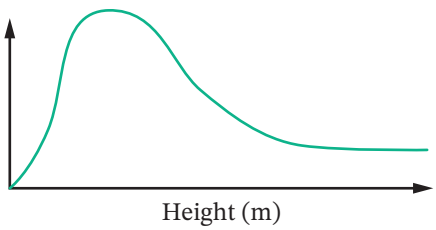

**b**



**c** Number of people waiting to catch a train




**d** Height of trees in a forest




**2** Which statement is true about this data set? Select the correct answer **A**, **B**, **C** or **D**.

**A** positively skewed and unimodal

**B** positively skewed and bimodal

**C** negatively skewed and unimodal

**D** negatively skewed and bimodal

| | Maths test results |
|---|---|
| 9 | 7 6 6 4 3 3 2 |
| 8 | 9 9 8 8 5 4 2 0 |
| 7 | 8 6 4 4 1 |
| 6 | 8 4 3 |
| 5 | 6 |

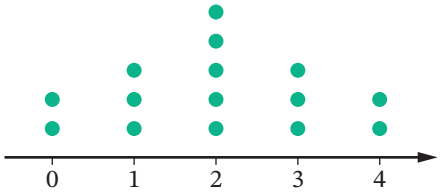
**3** The dot plot shown below shows the number of pets in each household in Lismore Ave. R C

**a** How many households were surveyed?

**b** Describe the shape of the data.

**c** What is the mode?

**d** Are the mean and median equal to the mode?




**4** The pulse rate (in beats/min) of the players of the Canberra football team is shown. R C

**a** Describe the shape of the data set.

**b** How many players are there in the team?

**c** Find the mean pulse rate for the team.

**d** Find the median pulse rate for the team.

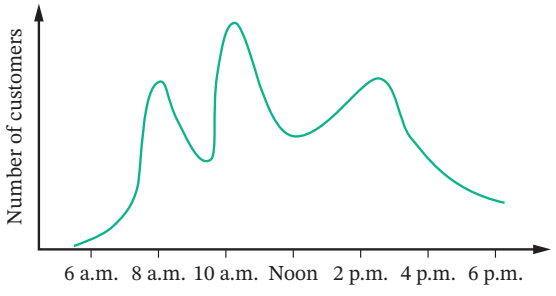
**e** Is the mean or median further away from the mode?

| Stem | Leaf |
|---|---|
| 5 | 0 8 |
| 6 | 4 8 |
| 7 | 0 4 6 |
| 8 | 2 2 2 6 6 6 |
| 9 | 0 0 0 0 0 0 2 4 |

**5** This 'trimodal' distribution represents the number of customers at a city business at different times of the day. R C

**a** Why is it called a trimodal distribution?

**b** What type of business do you think it could be? Give a reason for your answer.




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**6** This stem-and-leaf plot shows the exam marks of a group of History students. R C

| Stem | Leaf |
|---|---|
| 5 | 3 6 7 8 9 |
| 6 | 2 5 4 |
| 7 | 1 2 |
| 8 | 2 |
| 9 | 0 |

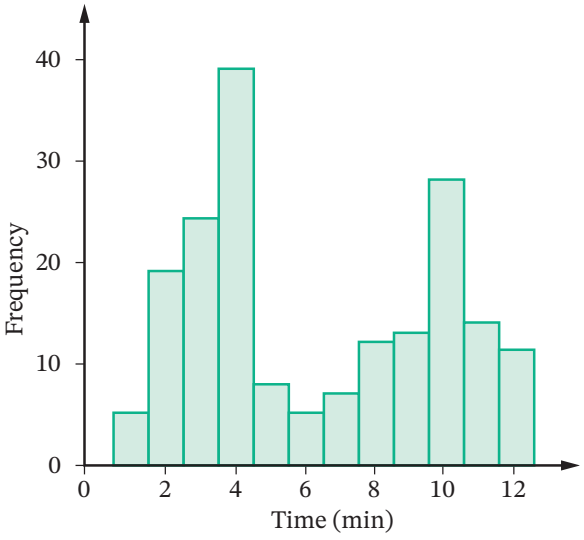
**a** Describe the shape of the distribution.

**b** Change 4 of the exam marks to make the distribution more like a symmetrical distribution. For example, you could change the 59 to 79.

**7** This histogram shows the amount of time customers at an appliance store had to wait for items to be brought in from the warehouse. R C

**a** Describe the shape of the distribution.

**b** Suggest a possible reason for the shape of the distribution.




**8** Sketch a graph to represent each distribution described. Remember to label the axes. R C

**a** The age at which a person dies is negatively skewed, with a mode of 80 years.

**b** The percentage marks students scored on a difficult exam has a bimodal, positively skewed distribution. A small group scored 90% but the majority scored around 40%.

**c** A bus carrying passengers to a concert includes a large group of dancers and a small group of rugby players. The distribution of the body massed (weights) of the passengers is bimodal.

**d** The heights of 2000 randomly-selected girls aged 17 are distributed symmetrically.

**e** When a die is rolled 100 times, the numbers that come up have a uniform distribution.

**f** The amount of time visitors spend at a museum exhibit that includes a 5-minute video is bimodal. Most visitors spend less than 1 minute looking at the exhibit, but some also watch the entire video and the exhibit. Very few people look at the display from 1 to 5 minutes.

# Comparing data

7.12

This table describes the 4 statistical measures and when it is more appropriate to use each one.

| Statistical measure | Features | When is it appropriate to use? |
|---|---|---|
| **Mean** | | |
| $\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$<br>For a frequency table:<br>$\bar{x} = \frac{\text{sum of } fx}{\text{sum of } f}$ | • For numerical data<br>• Depends on all the values in the data set<br>• Affected by outliers<br>• Can be a decimal that is not one of the actual values | When the data set does not have many outliers. |
| **Mode** | | |
| Most common value(s) | • For numerical and categorical data<br>• There may be more than one mode, or no mode at all<br>• Not affected by outliers | When the most common value or category is needed. |
| **Median** | | |
| Middle value, or average of the 2 middle values, when values are arranged in order | • For numerical data<br>• Can be one of the values<br>• Not affected by outliers | When the data set has many outliers. |
| **Range** | | |
| Highest value – lowest value | • For numerical data<br>• Depends on the highest and lowest values only | When a measure of spread is needed. |

## Example 19

A group of 8 children was surveyed about the amount of pocket money (in dollars) they received each week.

20 32 32 40 18 32 18 48

**a** Find:

**i** the mean **ii** the mode **iii** the median **iv** the range

**b** If the last student's pocket money increased from \$48 to \$75, describe how this would affect:

**i** the mean **ii** the mode **iii** the median **iv** the range

**c** Which measure of centre best represents the new set of data?

**Worksheets**
Analysing data
Comparing word lengths
Investigating young drivers

**Technologys**
Population polygons
Daily rainfall

7.12

## Solution

**a** **i** Mean: $\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$

$= \frac{20+32+32+40+18+32+18+48}{8}$

$= \frac{240}{8}$

$= \$30$

**ii** Mode $= \$32$

**iii** Rewriting the 8 values in order:

18 18 20 (32) (32) 32 40 48

There are 2 middle values, 32 and 32 (circled above).

Median $= \frac{32+32}{2}$

$= \$32$

**iv** Range $= 48 - 18$

$= \$30$

**b** If the last student's pocket money increased to \$75, this value would be a high outlier compared to the others.

**i** The mean should increase.

Mean: $\bar{x} = \frac{267}{8}$

$= \$33.375$

**ii** The mode will not change because \$32 is still the most common value.

**iii** The median will not change because the order of the values has not been affected.

18 18 20 (32) (32) 32 40 75

The 2 middle values are still 32 and 32 (circled above).

**iv** The range will increase because the highest value is now even higher.

Range $= 75 - 18$

$= \$57$

**c** The median is the best measure of centre because it is not affected by the outlier of \$75.

Note:

- The mode (\$32) is a good measure, but it only represents 3 of the values.
- The mean (\$33.375) is the least appropriate because it has been affected by the outlier and is now higher than most of the values.

**EXERCISE 7.12** ANSWERS ON P. 595

## Comparing data U F PS R C

In this exercise, round all mean answers to one decimal place where necessary.

EXAMPLE 19

7.12

**1** The distances (in kilometres) that a salesperson travelled each day for 10 days were:

49 77 63 56 49 40 56 40 74 49

**a** Find:

**i** the range **ii** the mode **iii** the median **iv** the mean.

**b** Which measure of centre best represents the data?

**2** The ages of the mathematics teachers at Westvale Secondary College are: R C

49 32 37 32 25 41 39

**a** Find:

**i** the range **ii** the mode **iii** the median **iv** the mean

**b** The 39-year-old teacher was replaced by a 22-year-old teacher. Without recalculating, describe how this will affect:

**i** the range **ii** the mode **iii** the median **iv** the mean

**3** In a swimming competition, each swimmer was graded A, B, C, D or E. One squad achieved the results shown. R C

| Grade | A | B | C | D | E |
|---|---|---|---|---|---|
| No. of swimmers | 4 | 9 | 8 | 6 | 3 |

**a** How many swimmers were in the squad?

**b** What is the mode?

**c** Why isn't it possible to find the mean or median of these grades?

**4** The annual incomes (× $1000) of 12 people are: R C

84 57 121 75 81 134 66 484 210 93 125 90

**a** Find (in dollars):

**i** the range **ii** the median **iii** the mean

**b** Why isn't the mean or mode an appropriate measure of centre for the incomes?

**c** Remove the outlier from the set of data and recalculate the mean and median. Are they closer together now?



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**5** The heights (in centimetres) of a team of basketball players are: R C

204 196 181 185 190 198 193

**a** Find:

**i** the mean **ii** the median **iii** the mode **iv** the range

**b** The 190 cm player is replaced by a 181 cm player. Without recalculating, describe how this will affect:

**i** the mean **ii** the median **iii** the mode **iv** the range

**6** The cricket batting scores for Sunjeet and Chris over six innings are shown. PS R C

Sunjeet: 29, 35, 40, 26, 28, 39 Chris: 2, 2, 3, 2, 188, 5

**a** What are the median scores for each player? Select the correct answer **A**, **B**, **C**, or **D**.

**A** Sunjeet 32, Chris 2 **B** Sunjeet 40, Chris 2.5

**C** Sunjeet 33, Chris 2.5 **D** Sunjeet 32, Chris 2.5

**b** Find the mean score for each player.

**c** Who is the better player if:

**i** the median is used? **ii** the mean is used?

**d** Who do you think is the better player? Explain your answer.

**7** Hardly Normal Discounts recorded the weekly number of sales of GPS navigators at 2 stores over a 20-week period. PS R C

*Forbes*: 44, 30, 54, 39, 63, 69, 75, 32, 80, 76, 90, 80, 55, 58, 71, 91, 36, 61, 65, 84

*Cobar*: 37, 36, 79, 77, 65, 47, 33, 49, 51, 60, 72, 49, 51, 73, 88, 31, 41, 97, 58, 30

**a** Display this information on a back-to-back stem-and-leaf plot.

**b** Calculate the mean, mode, median and range for each store.

**c** Which store performed better? Explain your answer.

**d** Which sales figure from the Cobar store is an outlier? Which of the measures of centre calculated in part **b** are most affected by this outlier?

**8** A netball coach kept a record of the number of goals scored by Christina and Mary during the first half of the season to compare their performances. PS R C

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Christina | 12 | 23 | 25 | 19 | 10 | 35 | 20 | 32 | 19 | 25 | 4 | 27 |
| Mary | 13 | 30 | 20 | 24 | 17 | 25 | 17 | 34 | 20 | 29 | 19 | 15 |

**a** Calculate the mean, mode, median and range for each player.

**b** Who do you think is the more consistent player? Explain.

**c** Who is the higher scorer?

**d** Who do you think is the better player? Why?

**e** What is the outlier value?

**f** If the outlier is removed, does it change:

**i** who the more consistent player is? **ii** who the higher scorer is?

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**9** 2 judges rated 8 contestants in a singing contest. R C

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Delta** | 8 | 5 | 6 | 8 | 5 | 9 | 2 | 6 |
| **Guy** | 7 | 8 | 5 | 4 | 9 | 5 | 5 | 6 |

**a** Draw a dot plot for each judge's marks.

**b** Calculate the range and median for each judge.

**c** Describe differences in the way each judge allocates marks.



## Power plus ANSWERS ON P. 595

**1** A small class achieved these marks in an exam:

66 68 74 76 82 79

**a** Find the mean of these marks, correct to one decimal place.

**b** The teacher realised there was an error in the marking and added 3 to each mark. Find the mean of the new marks.

**c** What effect did the extra 3 marks have on the mean?

**d** What effect does adding or subtracting the same number for all of the data have on the mean?

**2** Yelena sat 5 exams. Her average mark was 74%. What mark should Yelena obtain in her next exam if she wishes her average mark for the 6 exams to be 77%?

**3** If the mean of a set of values is 20, must 20 be one of the values? Give an example to illustrate your answer.

**4** 5 items of data were collected, but the figures were lost. The mean of the data was known to be 8 and the median was 9.

**a** What was the total of the data?

**b** What could the data have been?

**c** If you are told that the range is 7, what could the set of data have been?

**5** 4 sisters work at the same bank. One earns $500 per week, another earns $800 per week. The mean weekly wage of the 4 sisters is $2000. Is this possible? How?

# CHAPTER 7 REVIEW

## Language of maths

Quiz
Language of maths 7

Puzzle
Data crossword

| | | | |
|---|---|---|---|
| bias | categorical | census | cluster |
| continuous | discrete | dot plot | frequency histogram |
| frequency polygon | infographic | mean | measure of centre |
| median | mode | negatively skewed | nominal |
| numerical | ordinal | outlier | population |
| positively skewed | sample | stem-and-leaf plot | symmetrical |

**1** Name the **measures of centre**.

**2** What word means how often a value is listed in a set of data?

**3** What type of data is described by words rather than numbers?

**4** Is a **frequency polygon** a column graph or a line graph?

**5** What is a survey of the whole population called?

**6** What word describes a sample that misrepresents the population?

**7** List all the words that are types of graphs.

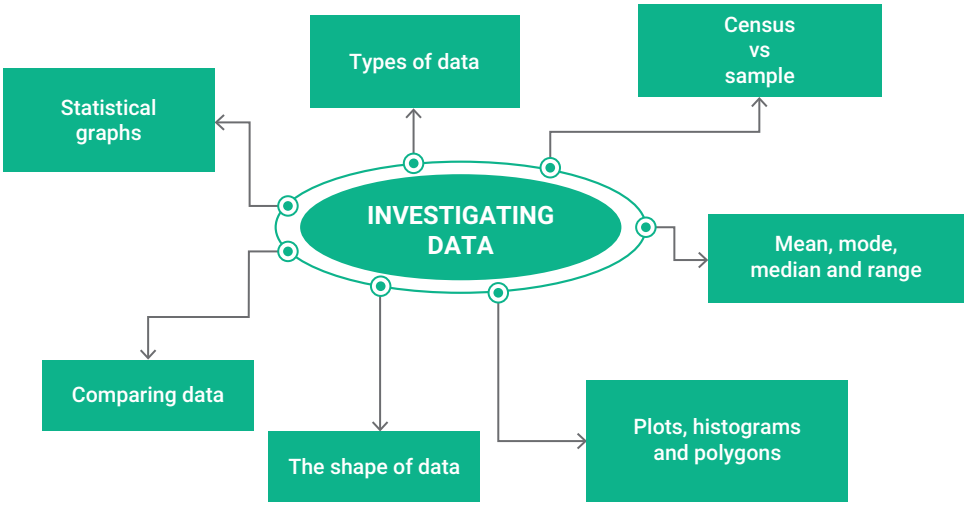
**8** List all the words that describe the shape of data.

## Topic summary

Worksheet
Mind map: Investigating data

- What did you find easy in this chapter? What don't you understand? Talk to your teacher.
- Give some examples of situations where statistics and measures of centre are used. Explain, where possible, which measure of centre is being used.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.





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# TEST YOURSELF 7

ANSWERS ON P. 595

Quiz
Test yourself 7



In this exercise, round all mean values to 2 decimal places where necessary.

**1** 20 families were asked how many computers they owned. 7.01

3 1 2 3 4 3 2 1 4 3
3 2 4 6 3 5 4 2 3 4

**a** Complete a frequency table for this data.
**b** What was the most frequent value?
**c** How many people owned 4 computers?
**d** What percentage of people owned one computer?
**e** Draw a sector graph of this data.

**2** Draw a pictogram for the average number of sunny days per month in Newcastle, shown below. 7.02

| J | F | M | A | M | J | J | A | S | O | N | D |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 12 | 13 | 13 | 17 | 13 | 16 | 18 | 18 | 16 | 13 | 15 |

**3** Classify each type of data below as categorical ordinal (CO), categorical nominal (CN), numerical discrete (ND) or numerical continuous (NC). 7.03

**a** life of a battery
**b** type of milk sold each day
**c** number of rooms in home
**d** volume of water in pools
**e** person's gender
**f** month of birthday

**4** Determine whether a sample or a census would be more appropriate for each survey below. 7.04

**a** Electing a Year 8 representative for student council.
**b** Investigating audience reaction to a new film.
**c** Testing the safety of a new car.
**d** Finding how many people live in Australia.

**5** A school of 1200 students wants to survey a sample of students to determine how often they use mobile phones. 7.04

**a** State a suitable way of choosing a representative sample of 50 students.
**b** Write one problem that may be encountered in conducting the survey.
**c** Explain how bias could be minimised.

**6** For each set of data, find: 7.05

**i** the mean **ii** the mode **iii** the median **iv** the range

**a** 6 12 11 12 10 6 6 10 6 7.06
**b** 8 4 1 1 4 1 3 6
**c** 20 21 23 24 25

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7.07

**7** The number of goals scored in each soccer match by the Southwest Strikers is shown in this frequency table.

| Score, $x$ | Frequency, $x$ | $fx$ |
|---|---|---|
| 0 | 3 | |
| 1 | 7 | |
| 2 | 6 | |
| 3 | 4 | |
| 4 | 2 | |
| 5 | 1 | |
| 6 | 1 | |
| Total | | |

**a** Copy and complete the frequency table.

**b** How many matches did the Strikers play?

**c** Find the range.

**d** Find the mode.

**e** Find the median.

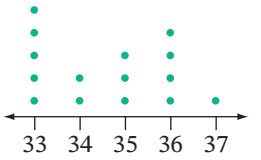
**f** Find the mean.

7.08

**8** For each plot below, find:

**i** the range **ii** the median **iii** the mode **iv** the mean

**a**




**b**

| Stem | Leaf |
|---|---|
| 12 | 3 5 6 |
| 13 | 1 1 |
| 14 | 0 8 8 8 8 |
| 15 | 7 8 |

7.09

**9** The daily number of people visiting a house that was for sale was:

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 5 | 6 | 5 | 4 | 5 | 4 | 3 | 5 |
| 3 | 5 | 6 | 6 | 5 | 4 | 3 | 3 | 5 | 4 |
| 1 | 2 | 2 | 5 | 4 | 3 | 6 | 6 | 2 | 5 |

**a** Arrange this information into a frequency table.

**b** Draw a frequency histogram and frequency polygon.

**c** On how many days did the house receive more than 4 visitors?

**d** Find the mode.

**e** Find the median.

7.10

**10** **a** What type of graph is represented by parallel rectangles separated by gaps?

**b** Name the 2 types of graph used to illustrate parts of a whole.

**c** What type of graph uses an icon or symbol to represent a numerical value?

**d** What type of graph usually has Time on the horizontal axis?

**e** What type of data is represented on a column graph?

**f** What type of plot lists the actual values of a data set?

**g** What type of plot is useful for small sets of numerical data?

**h** What type of graph is represented by rectangles not separated by gaps?

**i** Name one type of graph that easily shows the shape of a distribution.

**j** What type of graph uses joined lines to show the frequencies of numerical data?

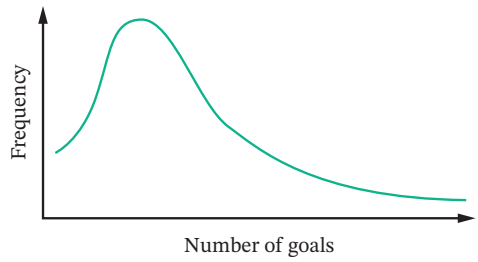
Foundation Standard Complex



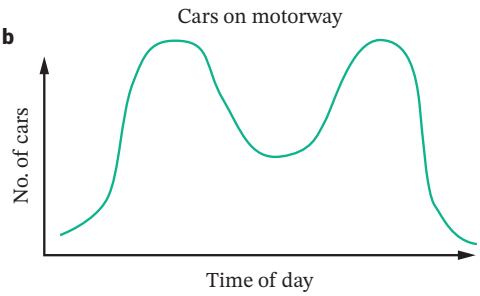
9780170479516

**11** What shape of distribution is shown in each graph? Choose from symmetrical, positively skewed, negatively skewed, uniform and bimodal. 7.11

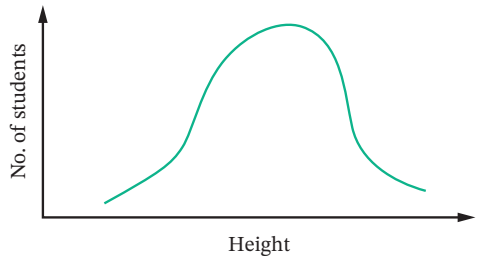
**a**




**b**




**c**




**12** Sketch a graph for each distribution described. Remember to label the axes. 7.11

**a** The percentage marks students scored on an easy test – the results are negatively skewed with most scoring between 80% and 90%.

**b** The masses of all 14-year-old girls in Australia are distributed symmetrically.

**c** When a playing die is rolled 500 times, the observed frequency of each value (1 to 6) has a uniform distribution.

**13** 21 Year 8 students were asked to find out the ages of their mothers. The results were: 7.12

30 35 35 35 36 36 36 36 36 37 37
37 37 38 39 39 41 43 43 45 45

**a** Find the median.

**b** Find the mode.

**c** Find the mean.

**d** Which measure of centre is the most appropriate for representing this data?

**e** Predict how many 39-year-old mothers there would be in a Year 8 group of 120 students.

**f** Suppose one of the ages is incorrect: one of the 39s should actually be 40. Without recalculating, describe how this will affect:

**i** the mean **ii** the median **iii** the mode **iv** the range